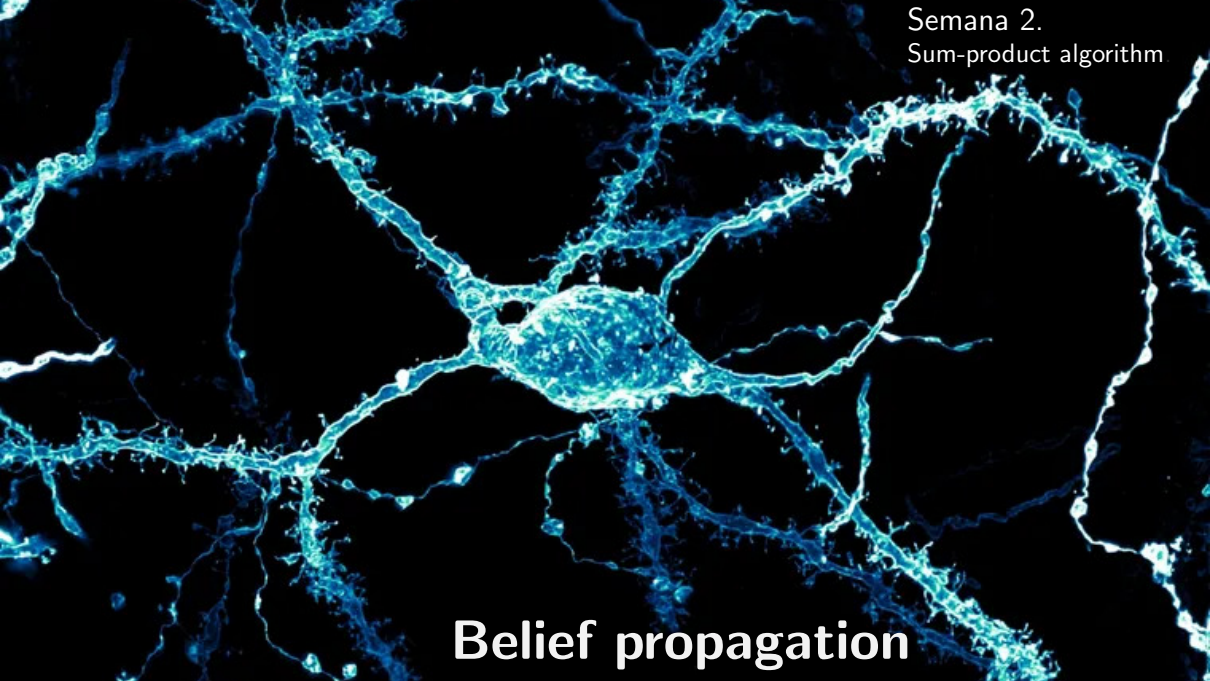


Semana 2.
Sum-product algorithm

Belief propagation



Bibliografía Semana 2

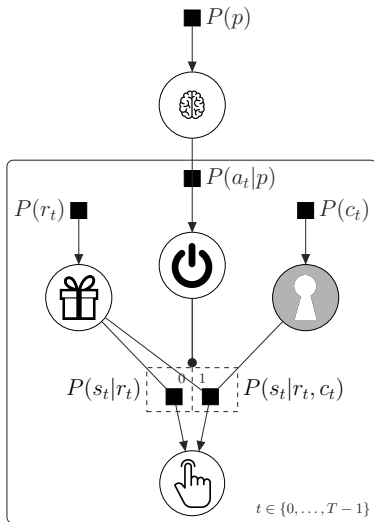
- Winn 2019. *Model-Based Machine Learning*. Cambridge university press; 2003 (Descargar). **Lectura: cap 1 y 2.**

Otras:

- Kschischang FR, Frey BJ, Loeliger HA. Factor graphs and the sum-product algorithm. 2001

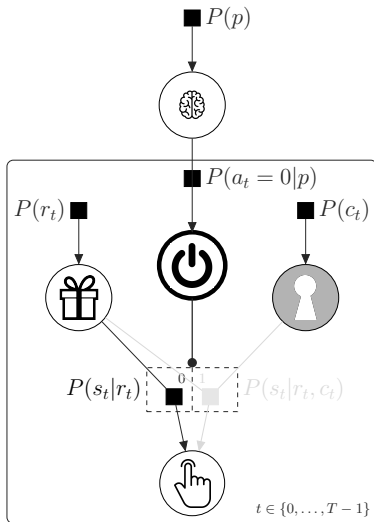
Mecanismos causales dinámicos

Gates



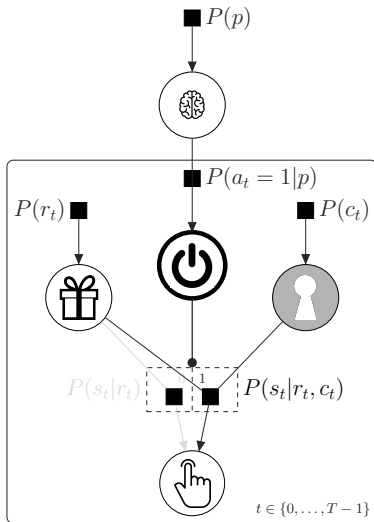
Mecanismos causales dinámicos

Gates



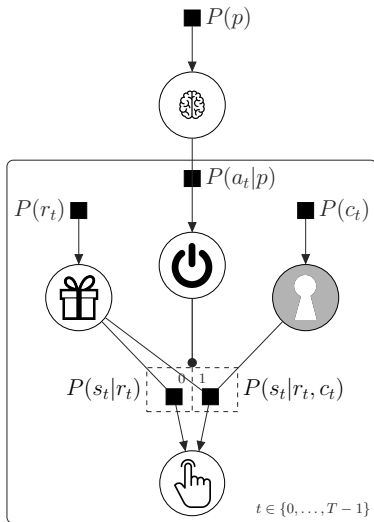
Mecanismos causales dinámicos

Gates



Mecanismos causales dinámicos

Gates

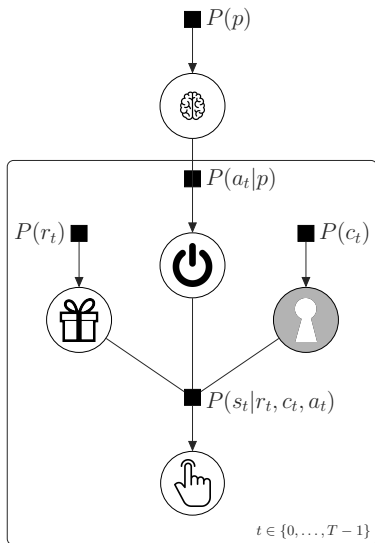


$$P(s_t|r_t, c_t, a_t) = P(s_t|r_t)^{\mathbb{I}(a_t=0)} P(s_t|r_t, c_t)^{\mathbb{I}(a_t=1)}$$

Flujo de inferencia

sum-product algorithm

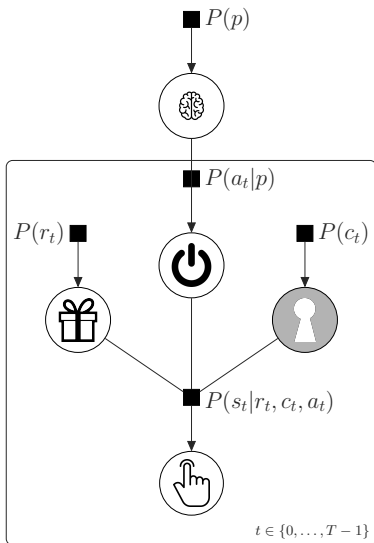
Algoritmo para calcular cualquier marginal
mediante pasaje de mensajes entre nodos



Flujo de inferencia

sum-product algorithm

Algoritmo para calcular cualquier marginal mediante pasaje de mensajes entre nodos



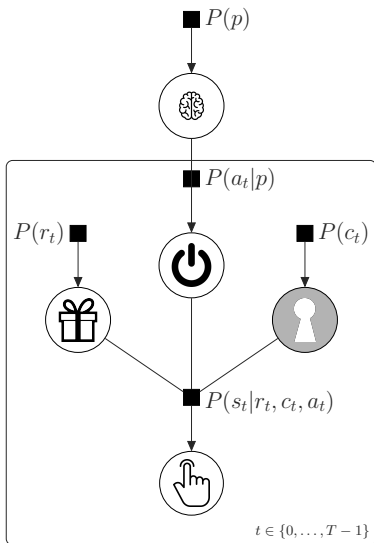
$m_{x \rightarrow f}(x)$: Mensaje de variable x a factor f

$m_{f \rightarrow x}(x)$: Mensaje de factor f a variable x

Flujo de inferencia

sum-product algorithm

Algoritmo para calcular cualquier marginal mediante pasaje de mensajes entre nodos



$v(n)$: Vecinos del nodo n

$m_{x \rightarrow f}(x)$: Mensaje de variable x a factor f

$m_{f \rightarrow x}(x)$: Mensaje de factor f a variable x

Flujo de inferencia

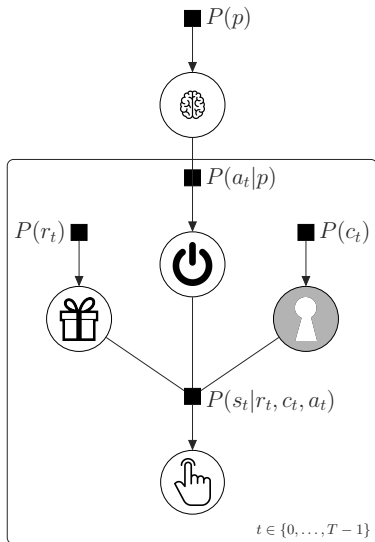
sum-product algorithm

$$P(x) = \prod_{f \in v(x)} m_{f \rightarrow x}(x)$$

$v(n)$: Vecinos del nodo n

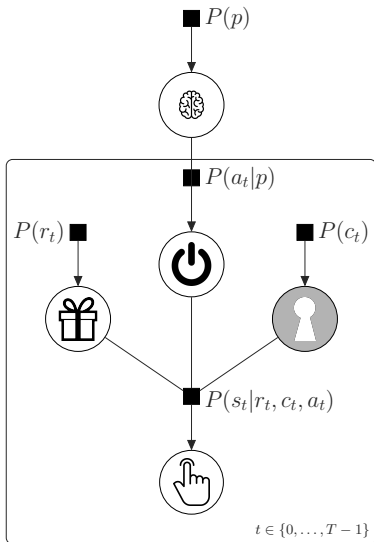
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Flujo de inferencia

sum-product algorithm



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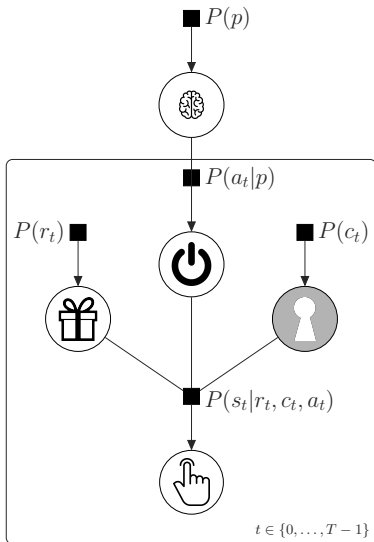
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$$m_{x \rightarrow f}(x) = \prod_{h \in v(x) \setminus \{f\}} m_{h \rightarrow x}(x)$$

Flujo de inferencia

sum-product algorithm



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$v(n)$: Vecinos del nodo n

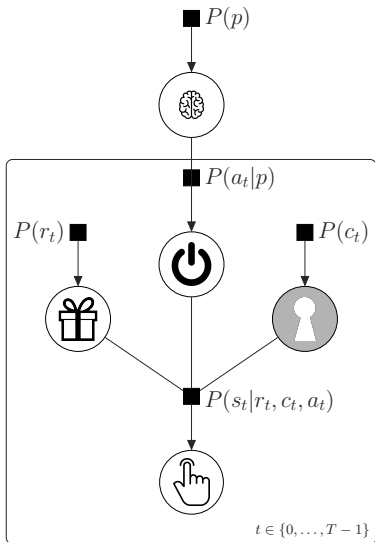
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Flujo de inferencia

sum-product algorithm



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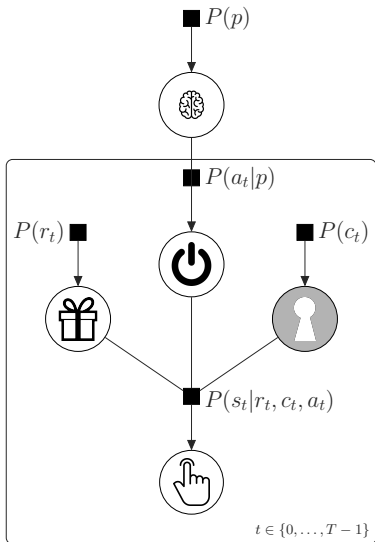
$m_{f \rightarrow x}(x)$: Mensaje de factor f a variable x

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Flujo de inferencia

sum-product algorithm



$$P(x) = \prod_{f \in v(x)} m_{f \rightarrow x}(x)$$

$v(n)$: Vecinos del nodo n

$m_{x \rightarrow f}(x)$: Mensaje de variable x a factor f

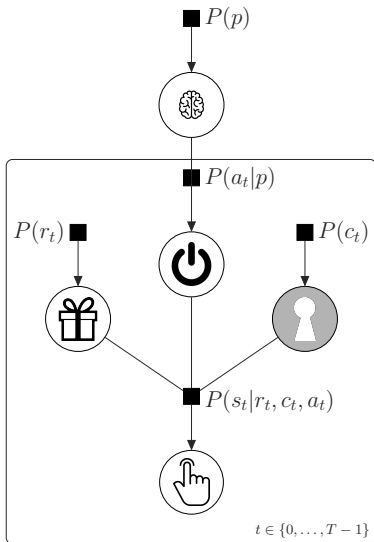
$m_{f \rightarrow x}(x)$: Mensaje de factor f a variable x

$$m_{x \rightarrow f}(x) = \prod_{h \in v(x) \setminus \{f\}} m_{h \rightarrow x}(x)$$

$$m_{f \rightarrow x}(x) = f(\mathbf{h}, x) \prod_{h \in v(f) \setminus \{x\}} m_{h \rightarrow f}(h)$$

Flujo de inferencia

sum-product algorithm



$$P(x) = \prod_{f \in v(x)} m_{f \rightarrow x}(x)$$

$v(n)$: Vecinos del nodo n

$m_{x \rightarrow f}(x)$: Mensaje de variable x a factor f

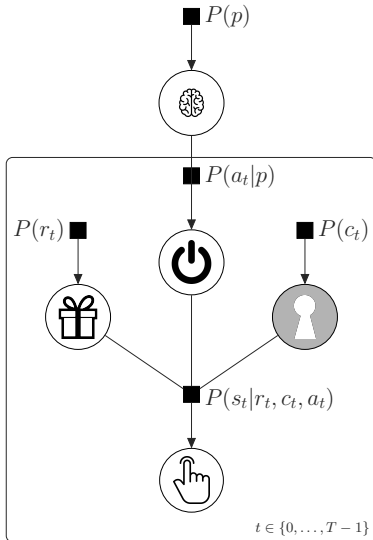
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$$m_{f \rightarrow x}(x) = \sum_{\mathbf{h}} \left(f(\mathbf{h}, x) \prod_{h \in v(f) \setminus \{x\}} m_{h \rightarrow f}(h) \right)$$

Flujo de inferencia

sum-product algorithm



$$P(x) = \prod_{f \in v(x)} m_{f \rightarrow x}(x)$$

$v(n)$: Vecinos del nodo n

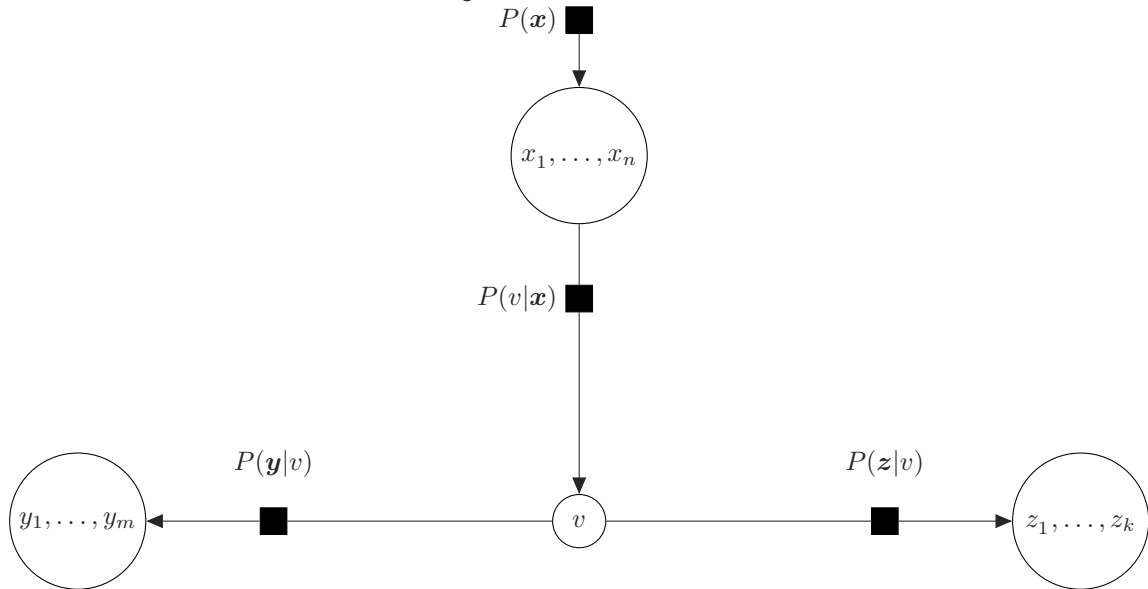
$m_{x \rightarrow f}(x)$: Mensaje de variable x a factor f

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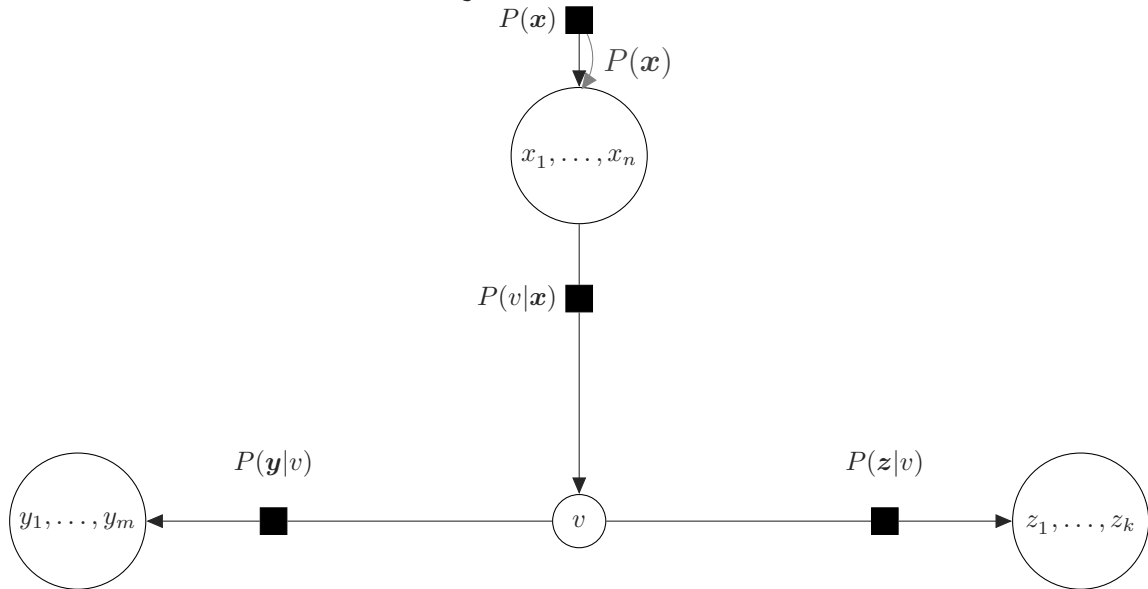
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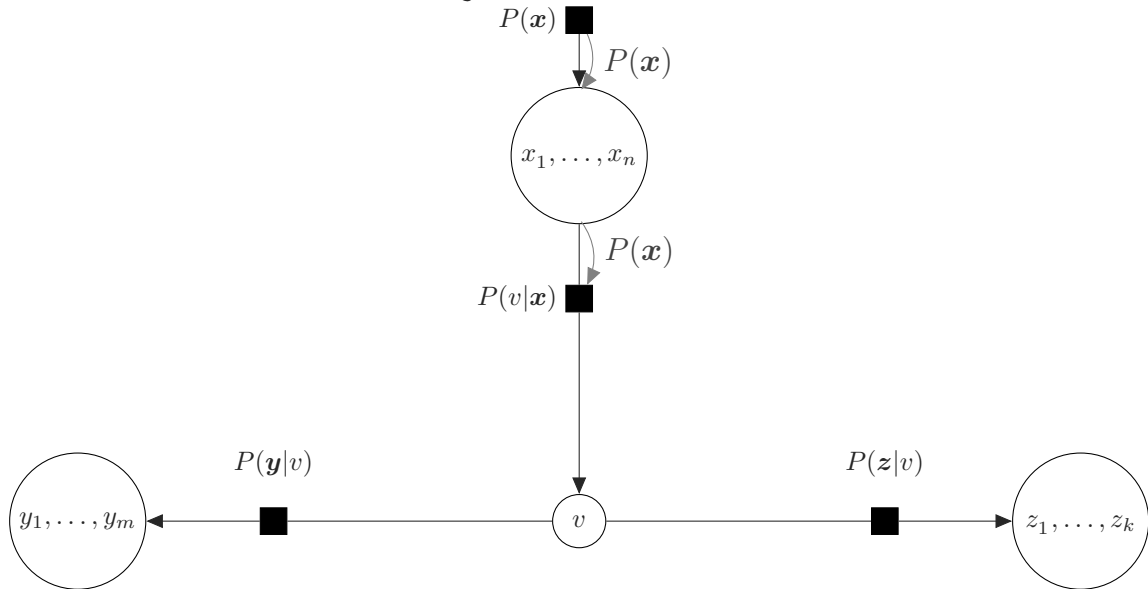
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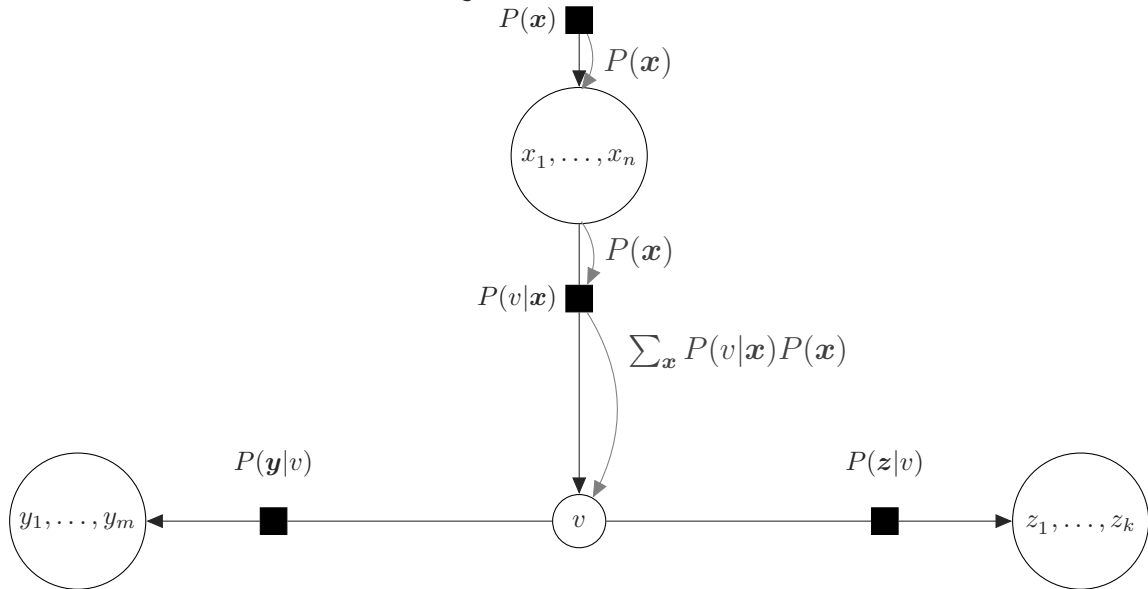
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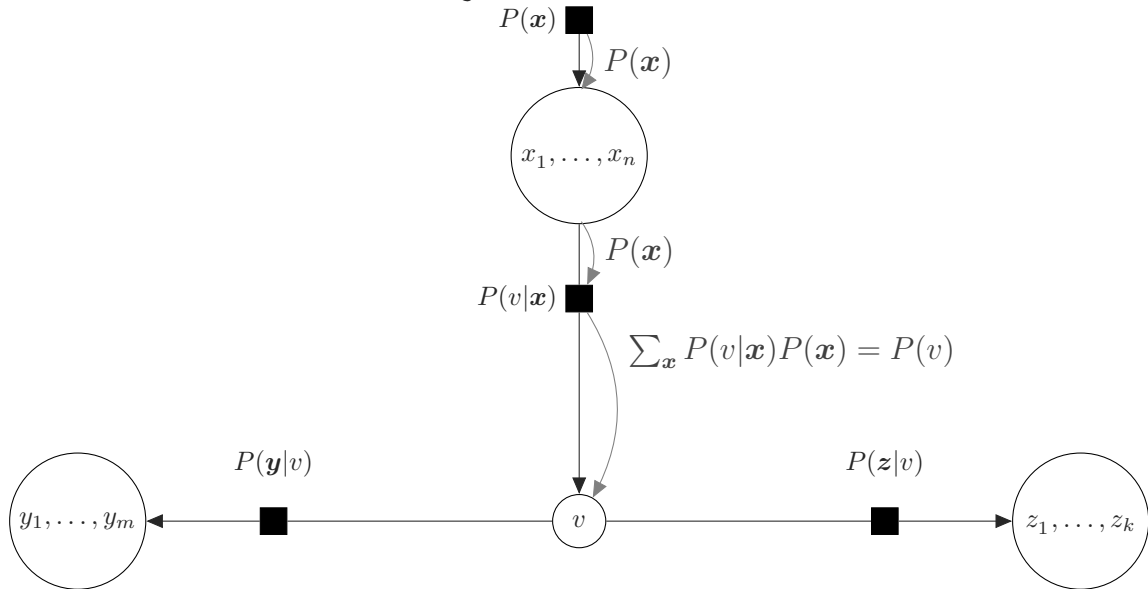
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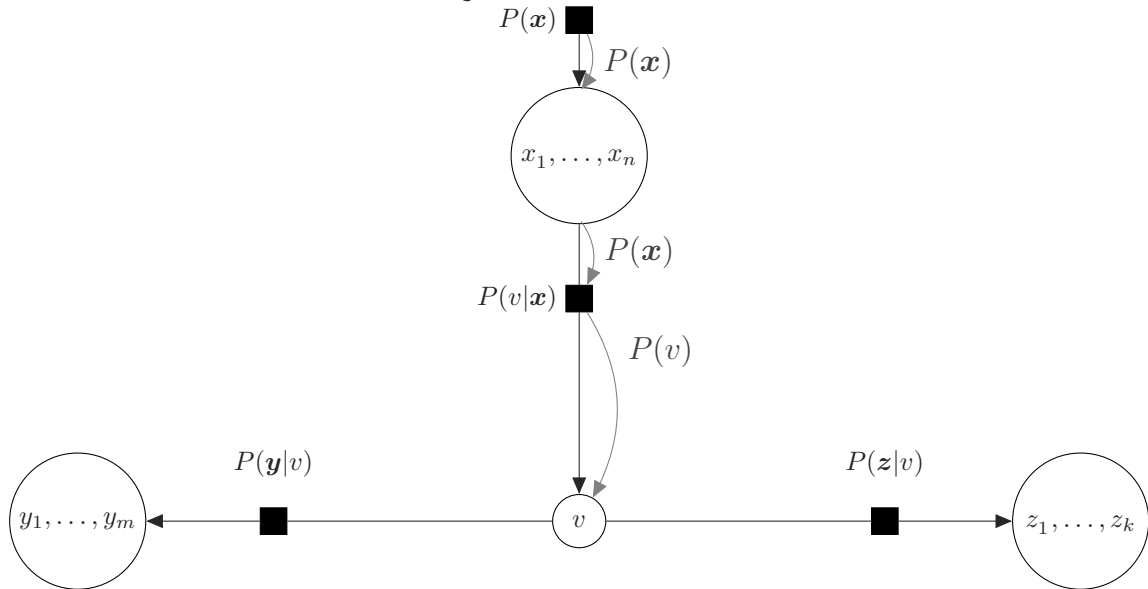
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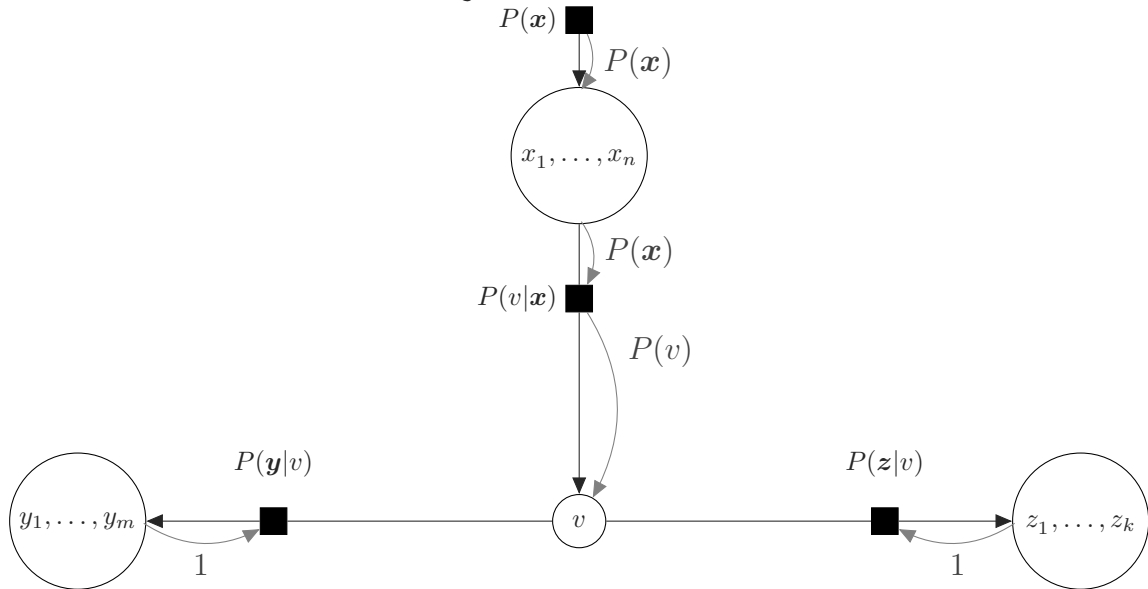
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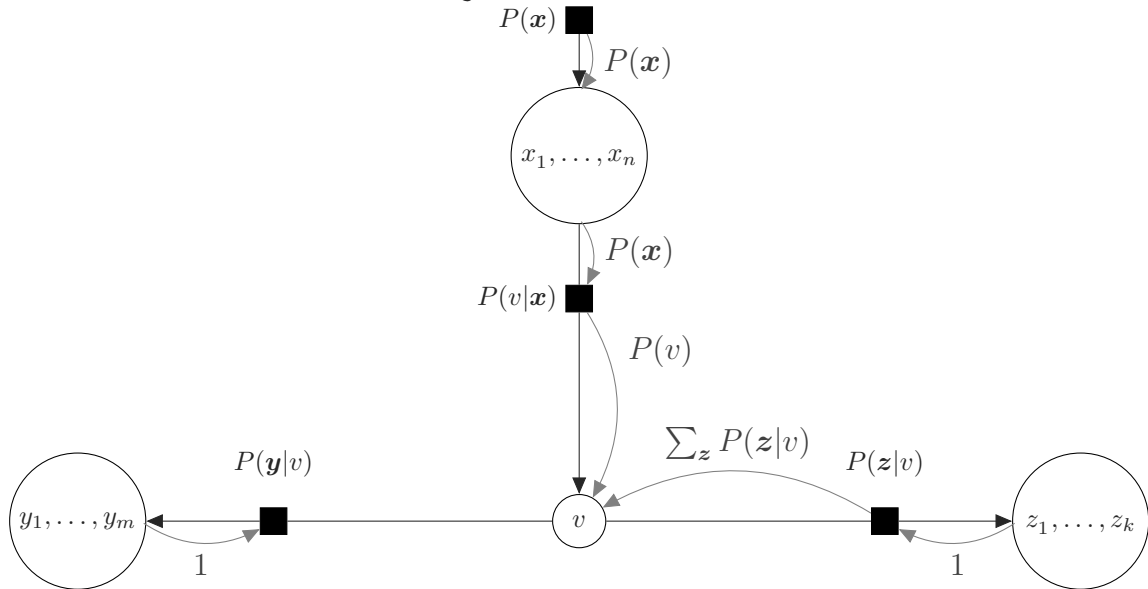
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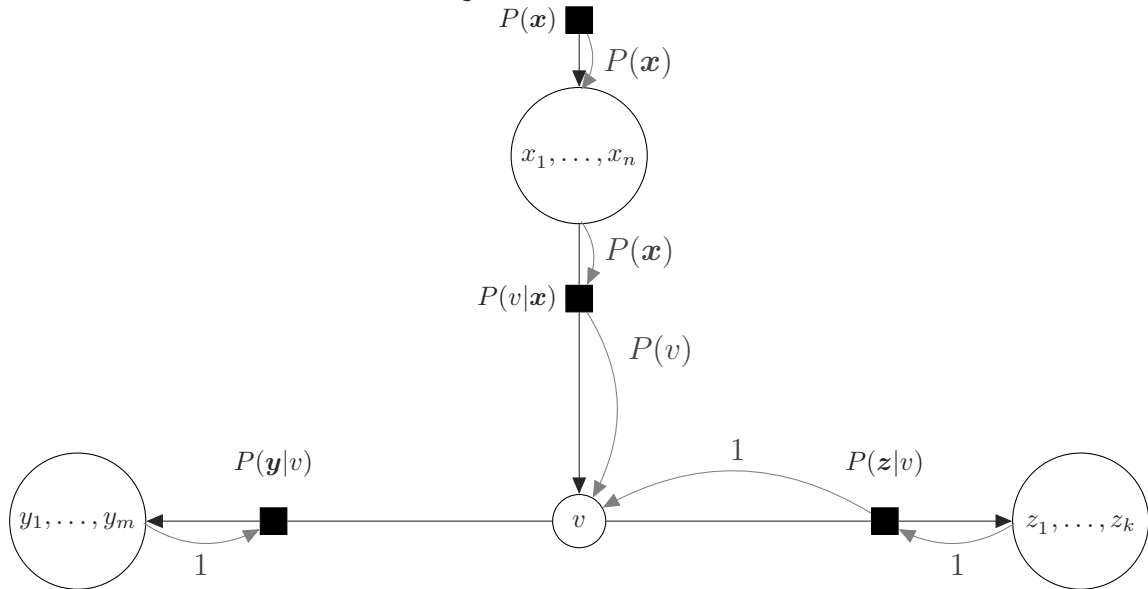
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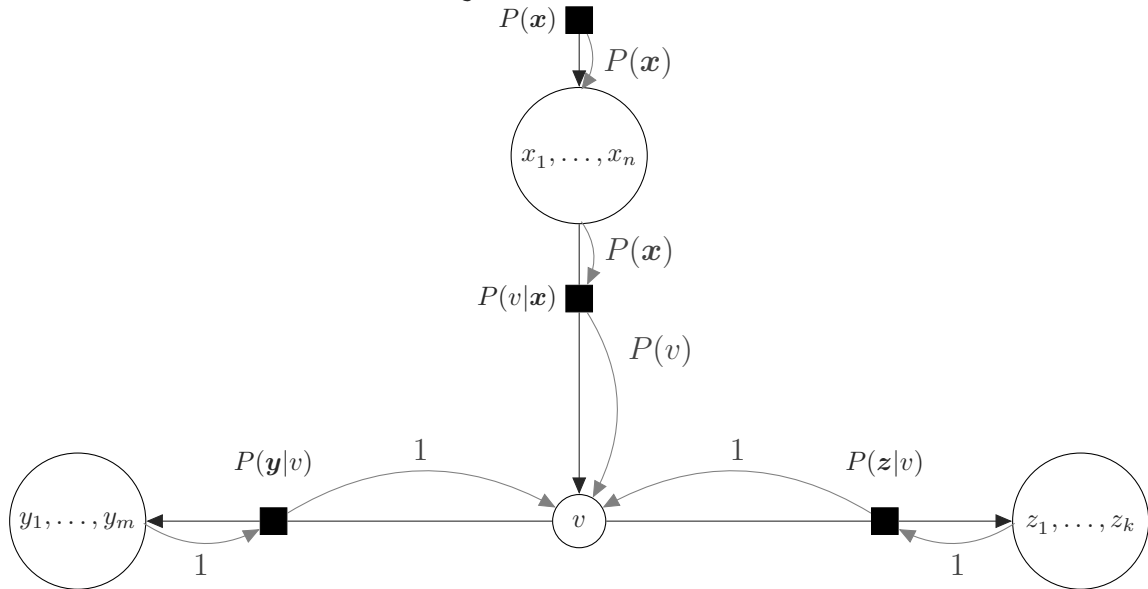
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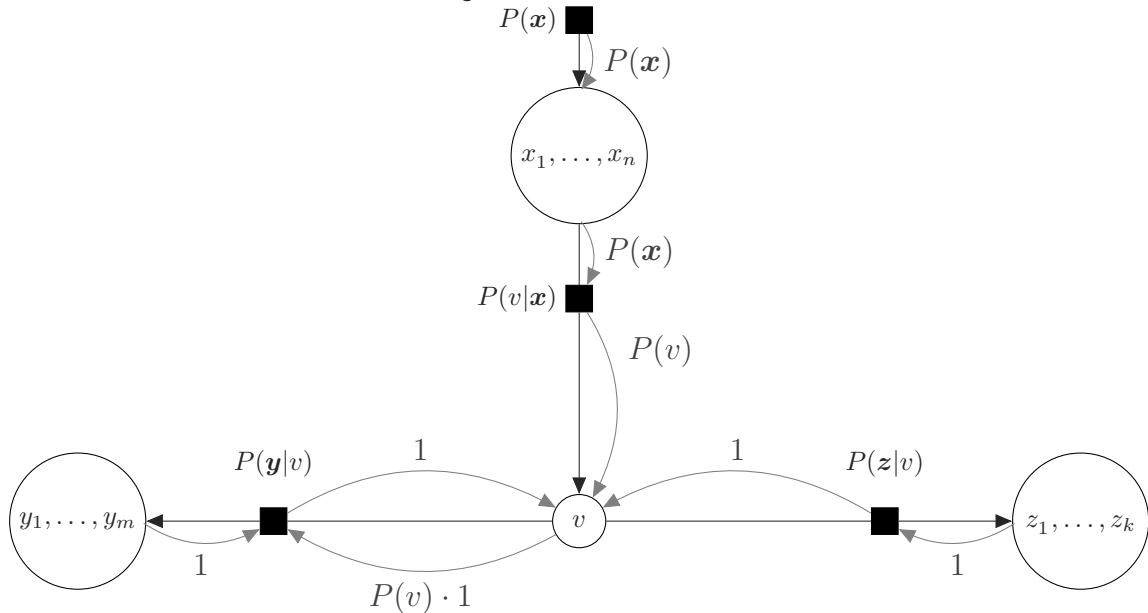
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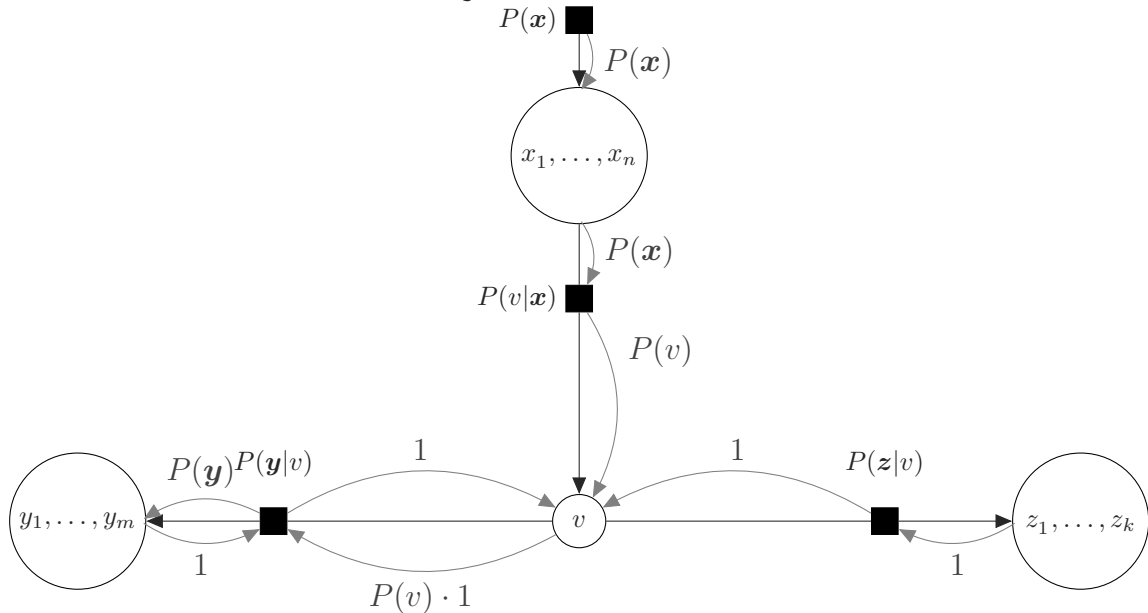
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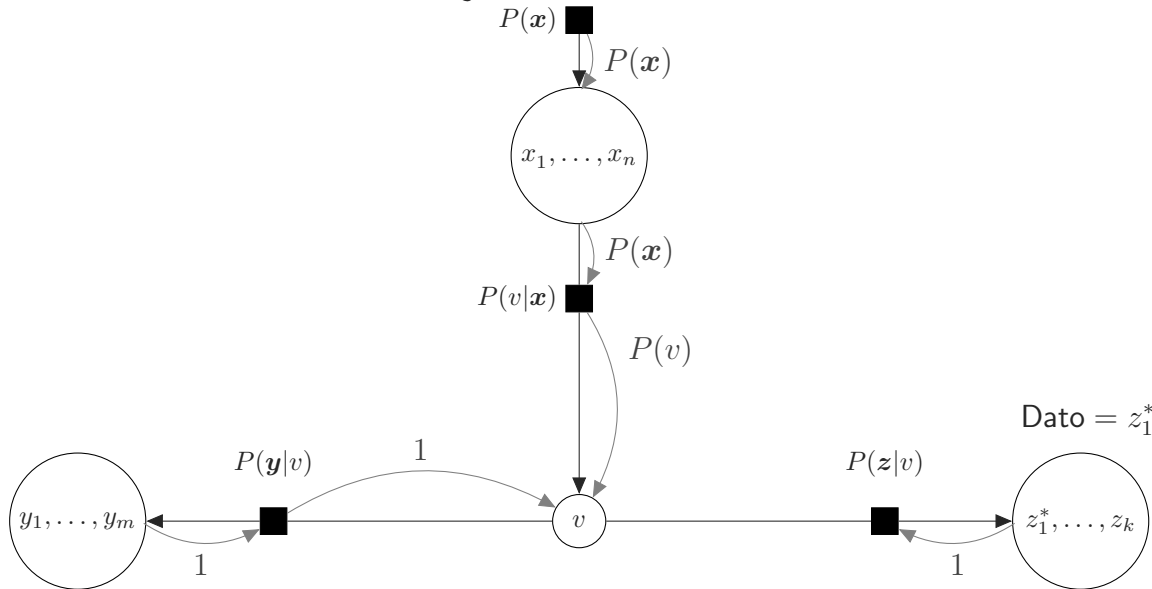
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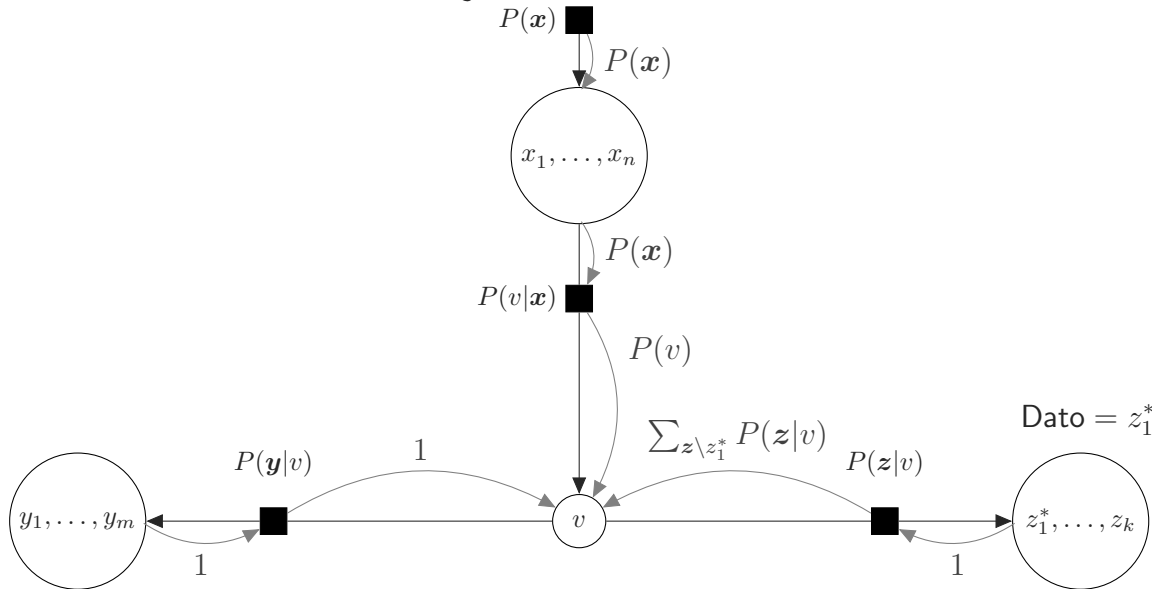
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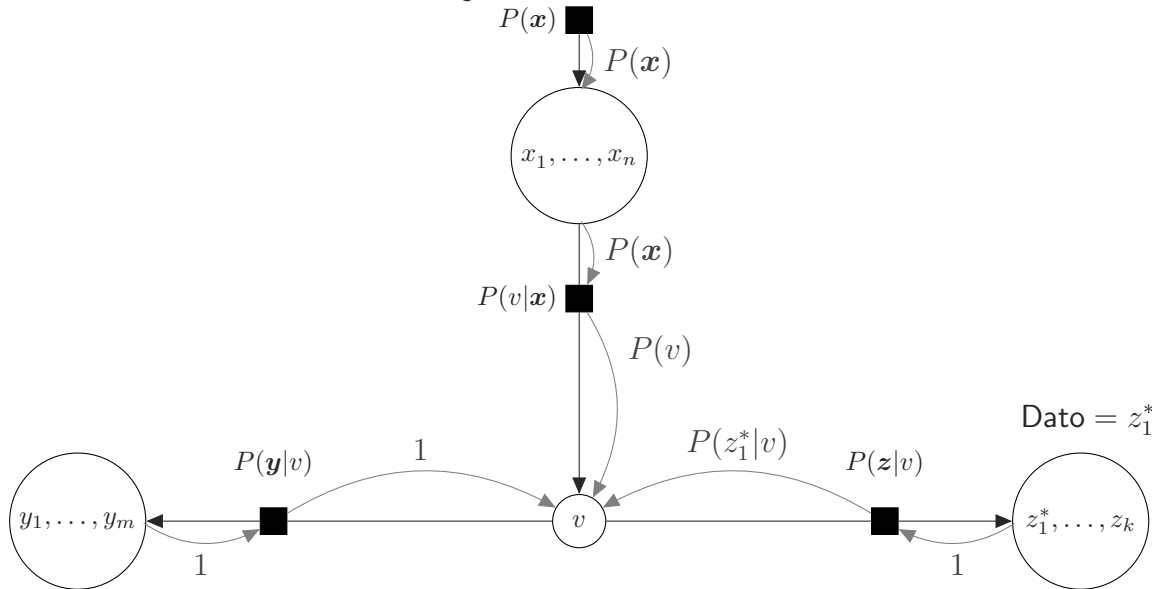
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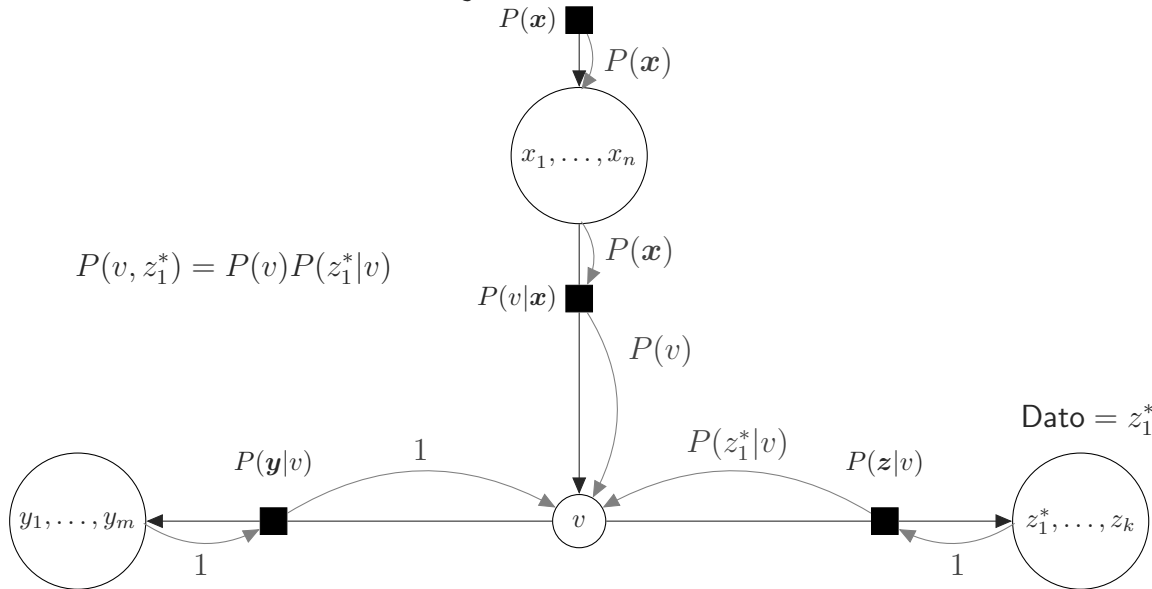
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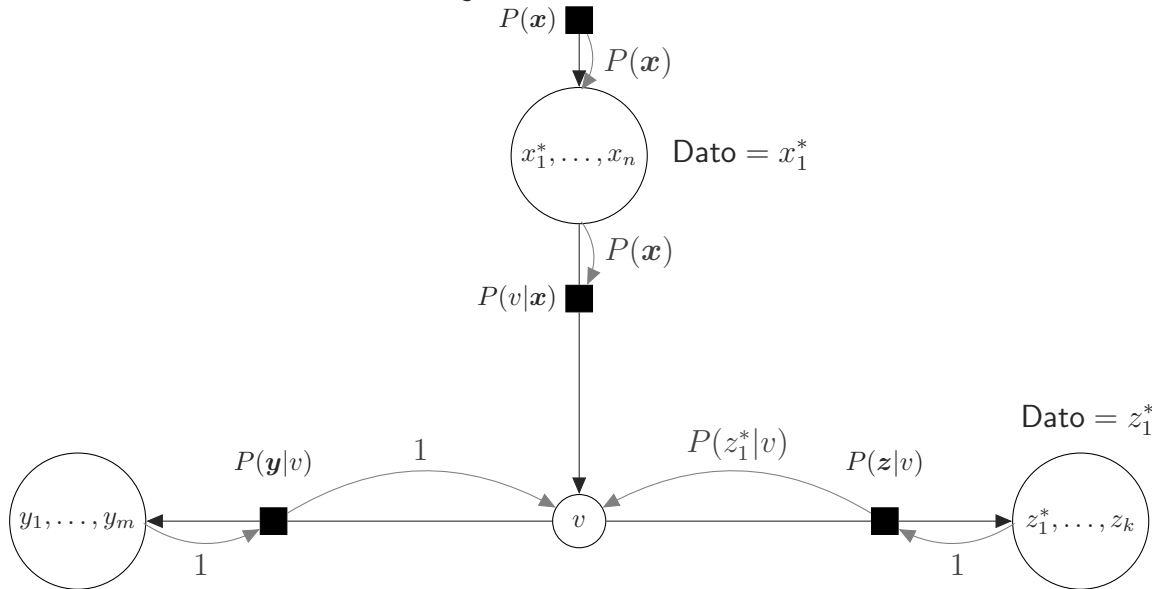
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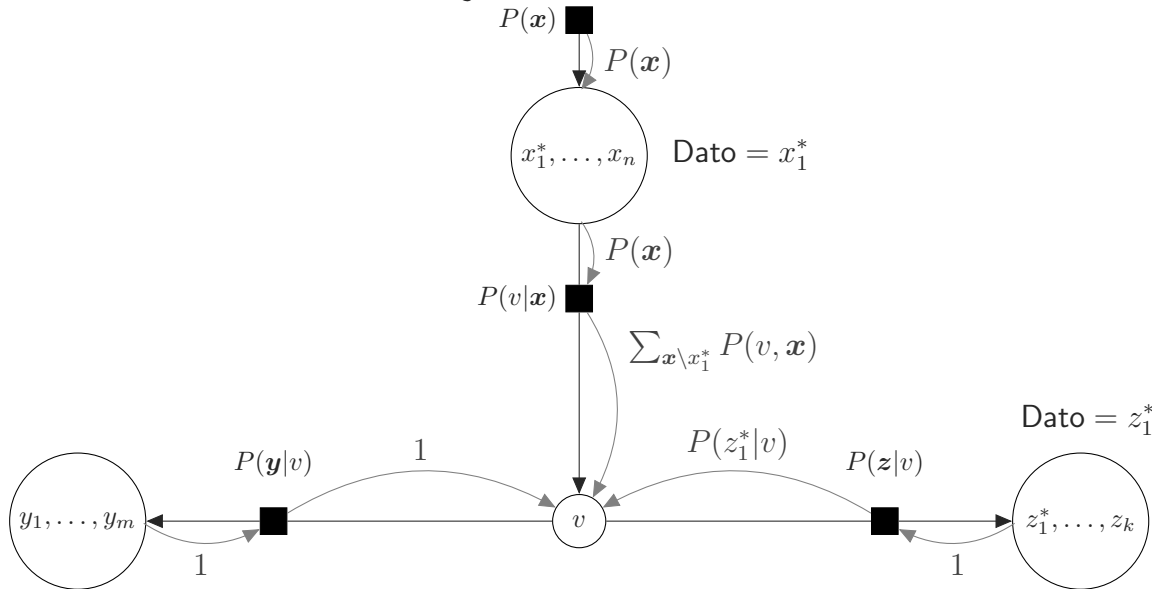
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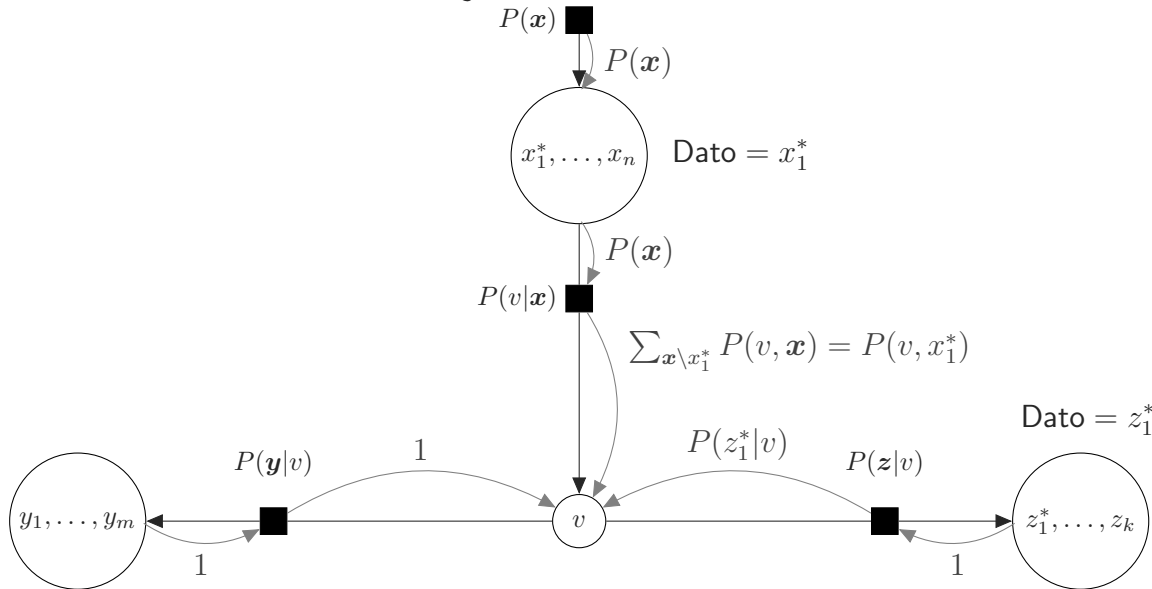
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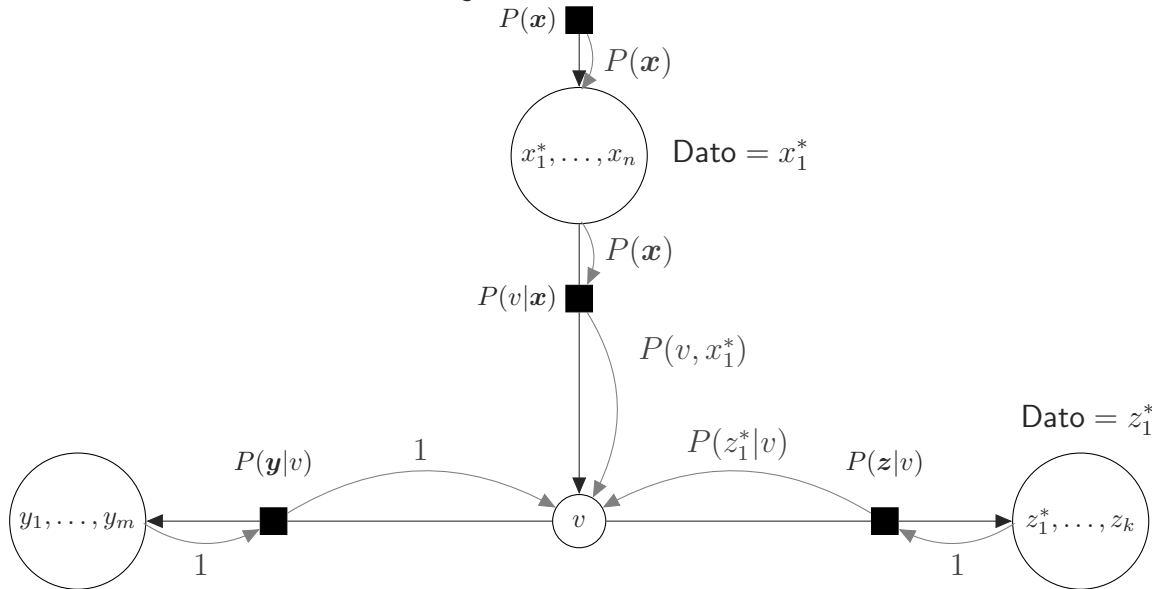
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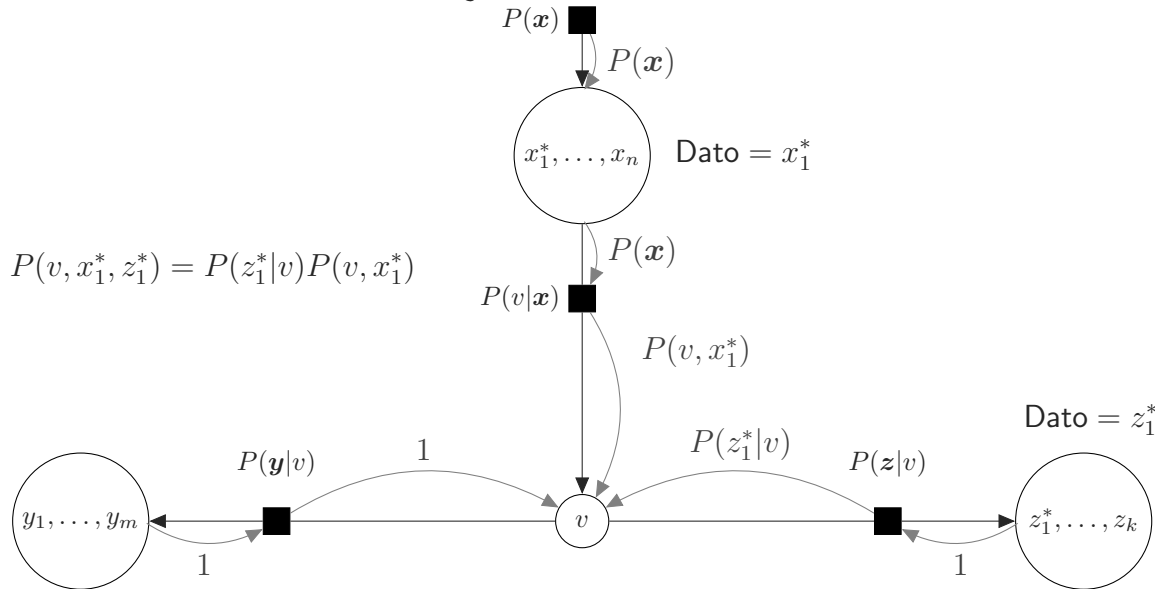
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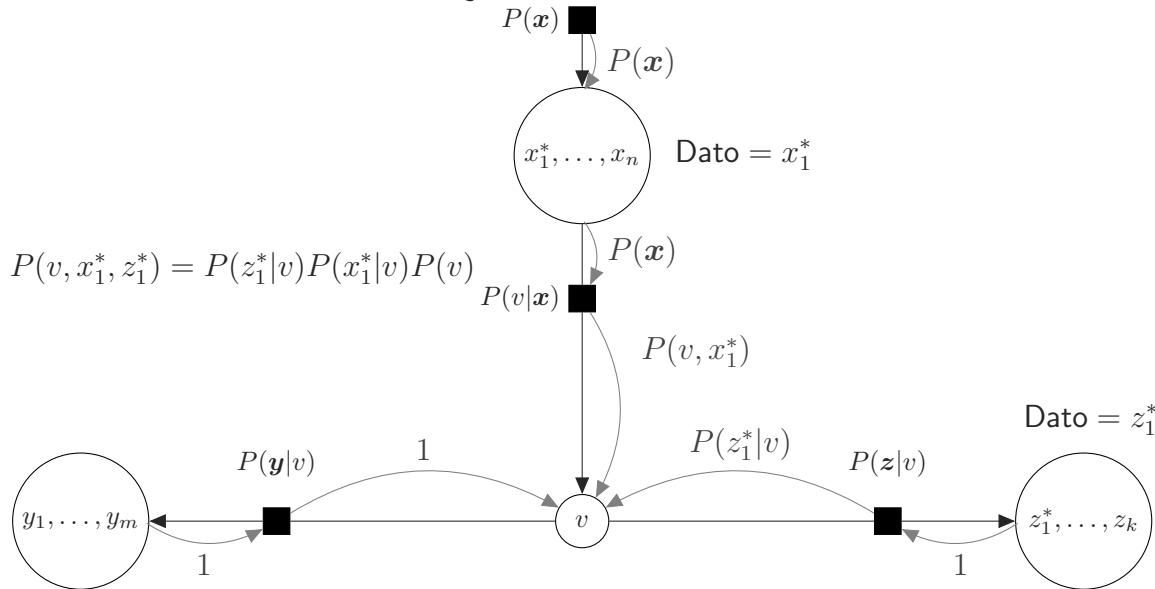
Flujo de inferencia



Flujo de inferencia

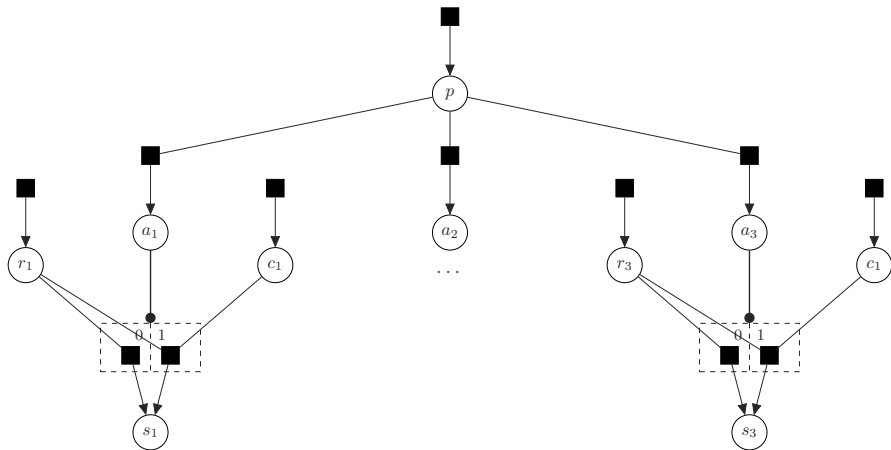


Flujo de inferencia



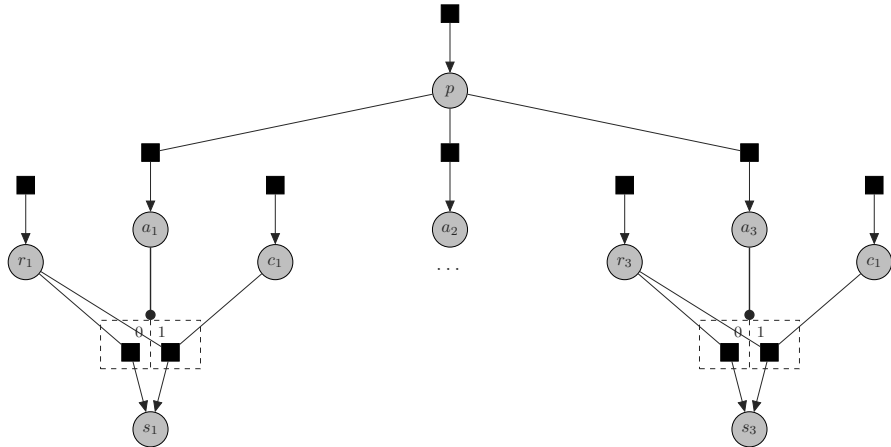
Flujo de inferencia

Todas las variables ocultas

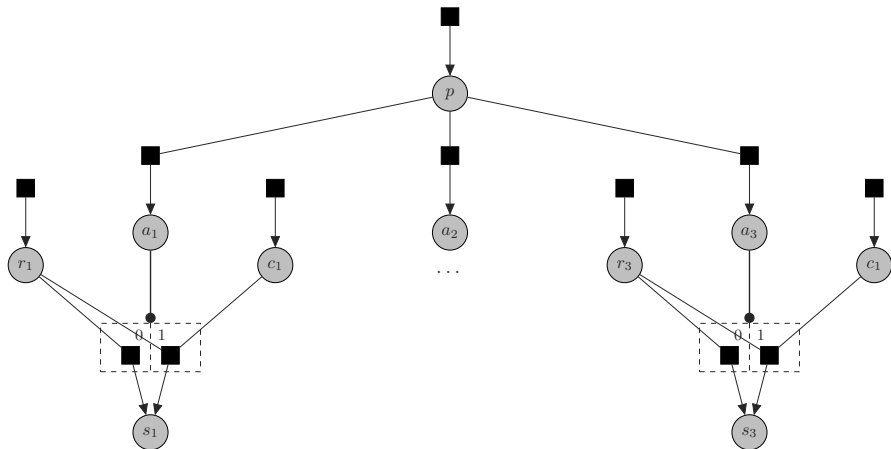


Flujo de inferencia

Todas las variables observadas

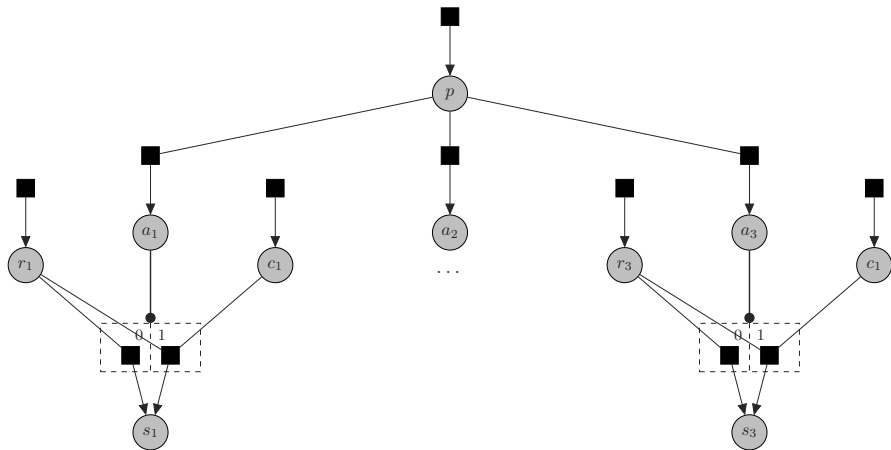


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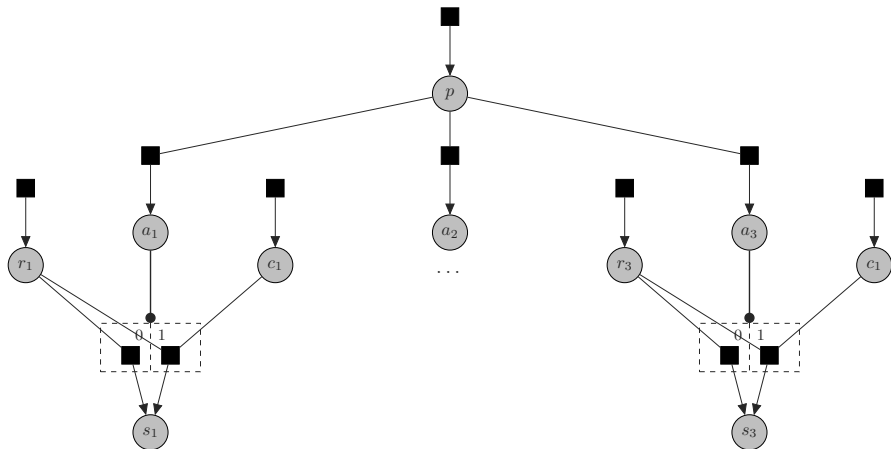
Recordar que las variables conocidas (observadas) están fijas (no se integran).

Flujo de inferencia



En este caso se van a ir acumulando el producto de distribuciones condicionales.

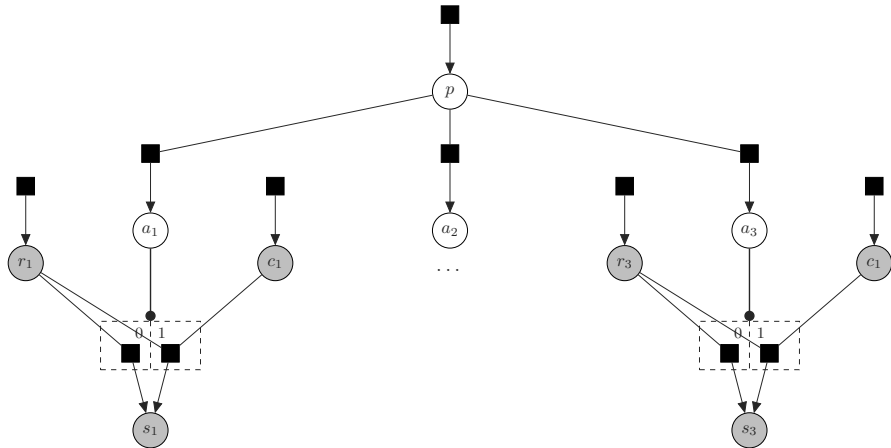
Flujo de inferencia



En este caso se van a ir acumulando el producto de distribuciones condicionales.
por lo que el producto de mensajes que llegan a una variable reconstruyen la distribución
conjunta

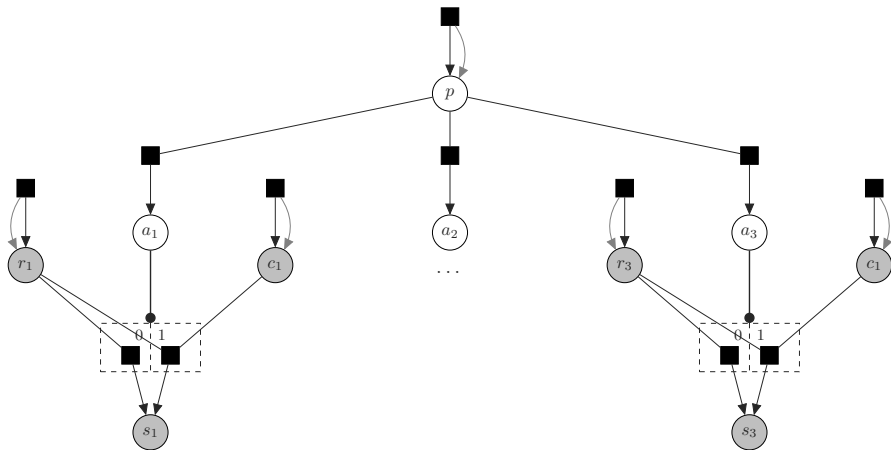
Flujo de inferencia

estructuras causales dinámicas



Flujo de inferencia

estructuras causales dinámicas



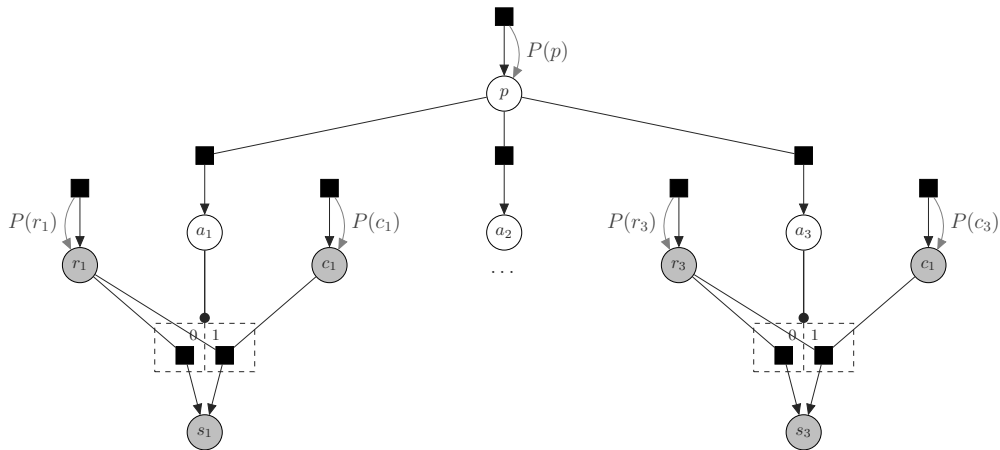
$$m_{f_r \rightarrow r}(r) =$$

$$m_{f_p \rightarrow p}(p) =$$

$$m_{f_c \rightarrow c}(c) =$$

Flujo de inferencia

estructuras causales dinámicas



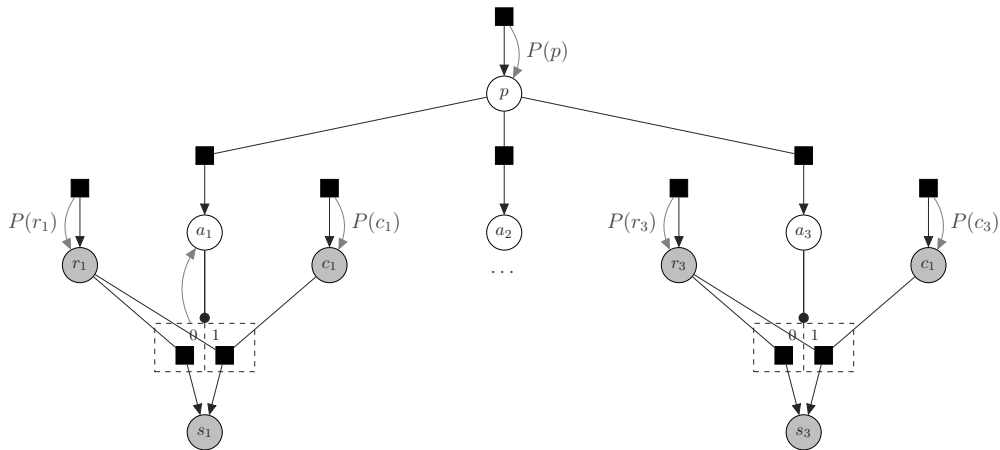
$$m_{f_r \rightarrow r}(r) = P(r)$$

$$m_{f_p \rightarrow p}(p) = P(p)$$

$$m_{f_c \rightarrow c}(c) = P(c)$$

Flujo de inferencia

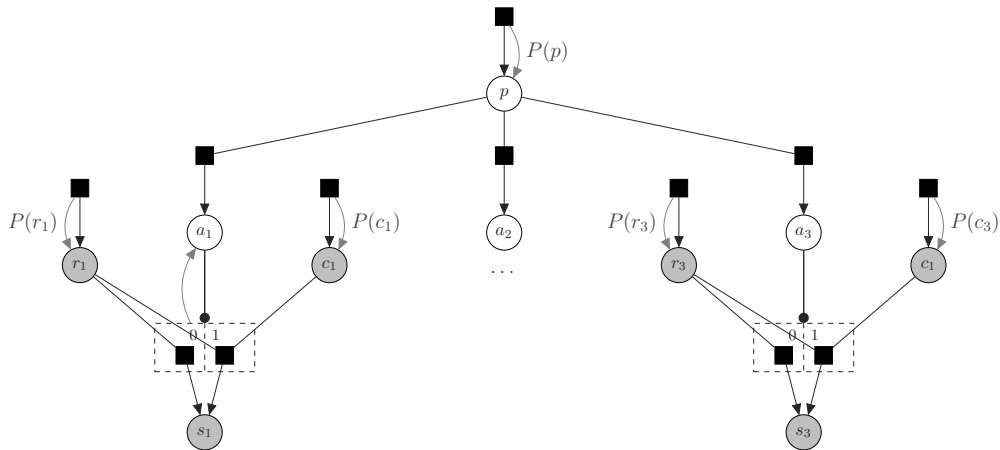
estructuras causales dinámicas



$$m_{f_s \rightarrow a}(a) =$$

Flujo de inferencia

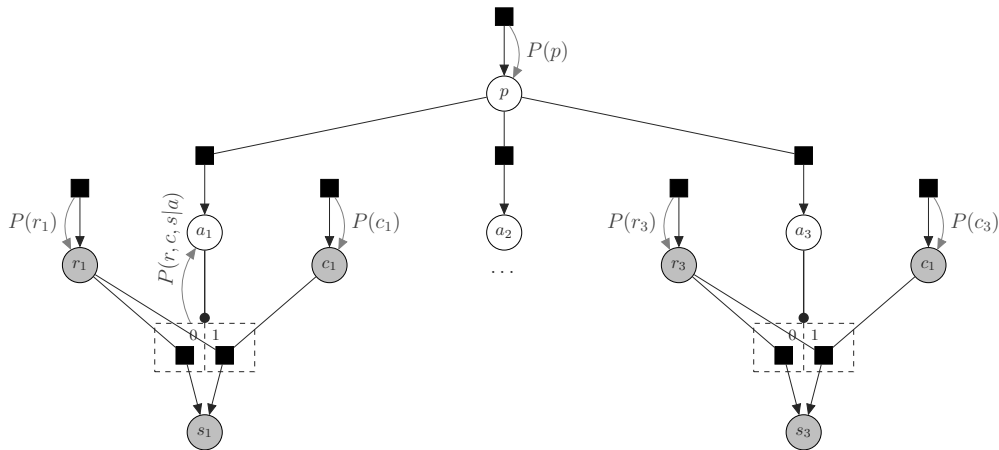
estructuras causales dinámicas



$$m_{f_s \rightarrow a}(a) = P(r)P(c)P(s|r)^{\mathbb{I}(a=0)}P(s|r, c)^{\mathbb{I}(a=1)}$$

Flujo de inferencia

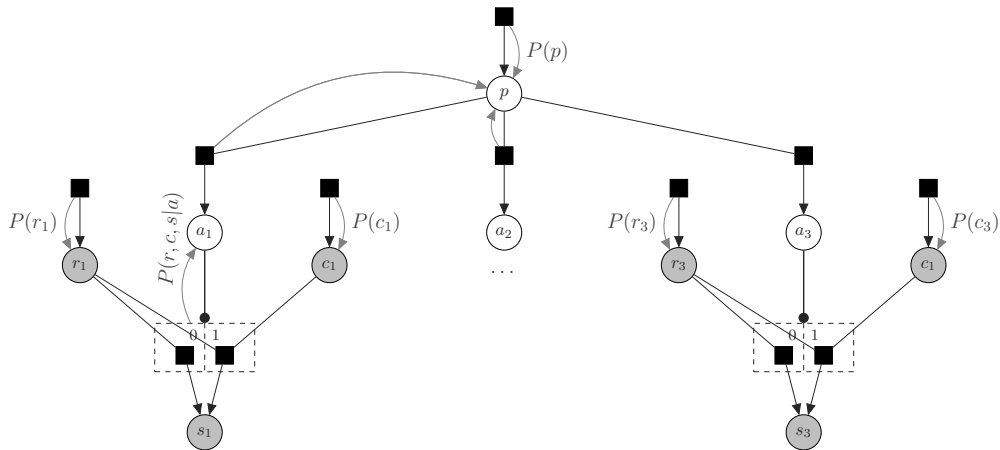
estructuras causales dinámicas



$$m_{f_s \rightarrow a}(a) = P(r)P(c)P(s|r)^{\mathbb{I}(a=0)}P(s|r, c)^{\mathbb{I}(a=1)} = P(r, c, s|a)$$

Flujo de inferencia

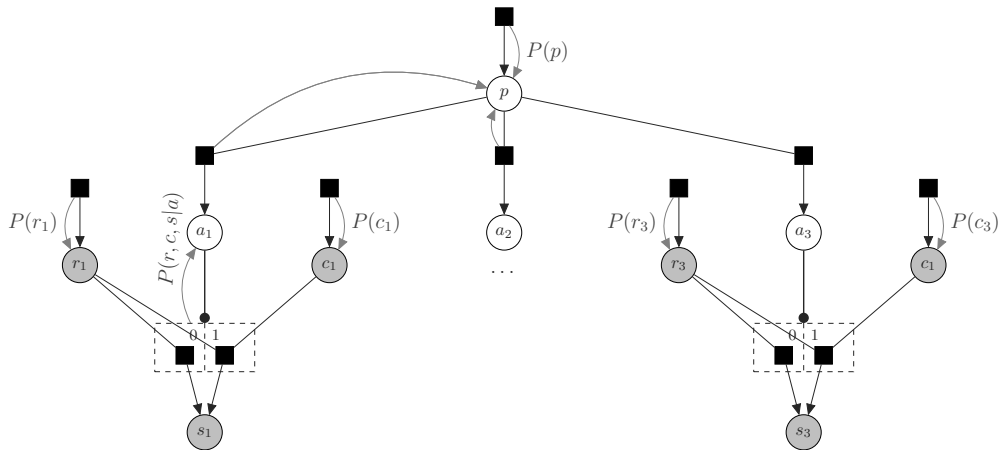
estructuras causales dinámicas



$$m_{f_a \rightarrow p}(p) =$$

Flujo de inferencia

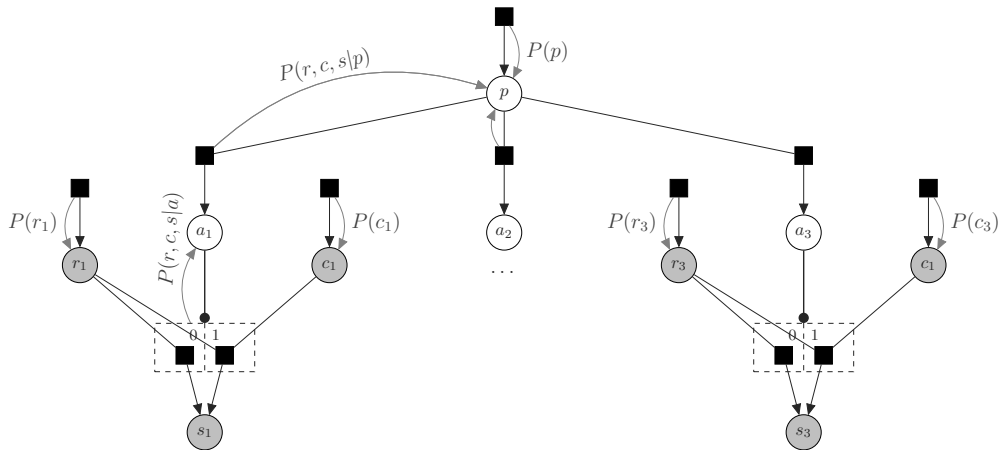
estructuras causales dinámicas



$$m_{f_a \rightarrow p}(p) = \sum_a P(r, c, s|a)P(a|p)$$

Flujo de inferencia

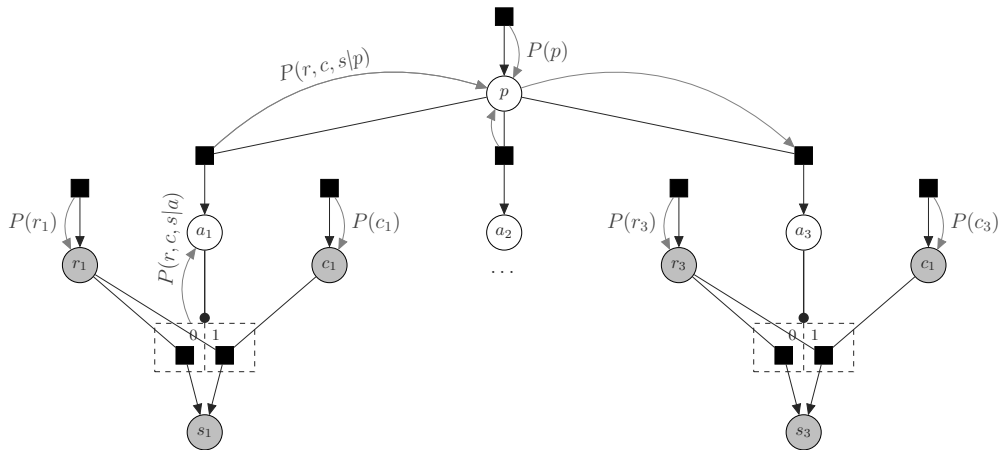
estructuras causales dinámicas



$$m_{f_a \rightarrow p}(p) = \sum_a P(r, c, s|a)P(a|p) = P(r, c, s|p)$$

Flujo de inferencia

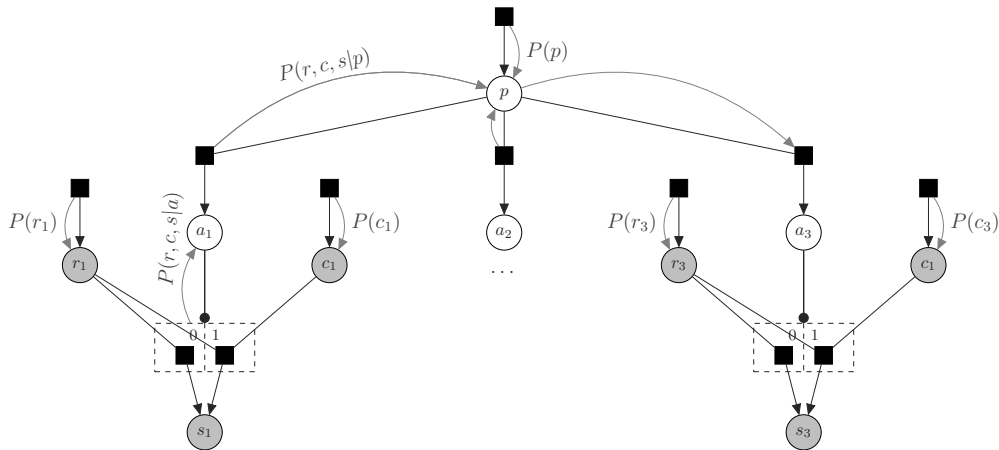
estructuras causales dinámicas



$$m_{p \rightarrow f_{a_{T+1}}}(p) =$$

Flujo de inferencia

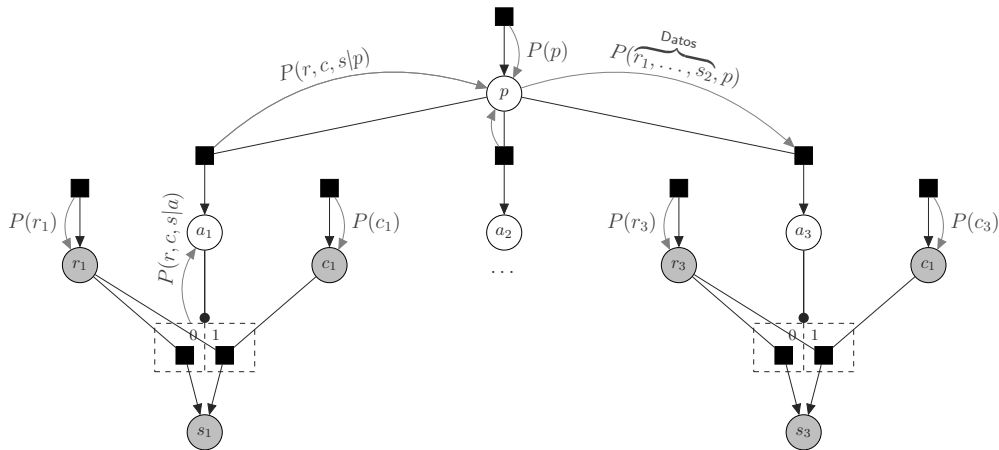
estructuras causales dinámicas



$$m_{p \rightarrow f_{a_{T+1}}}(p) = P(p) \prod_{i=1}^T P(r_i, c_i, s_i | p)$$

Flujo de inferencia

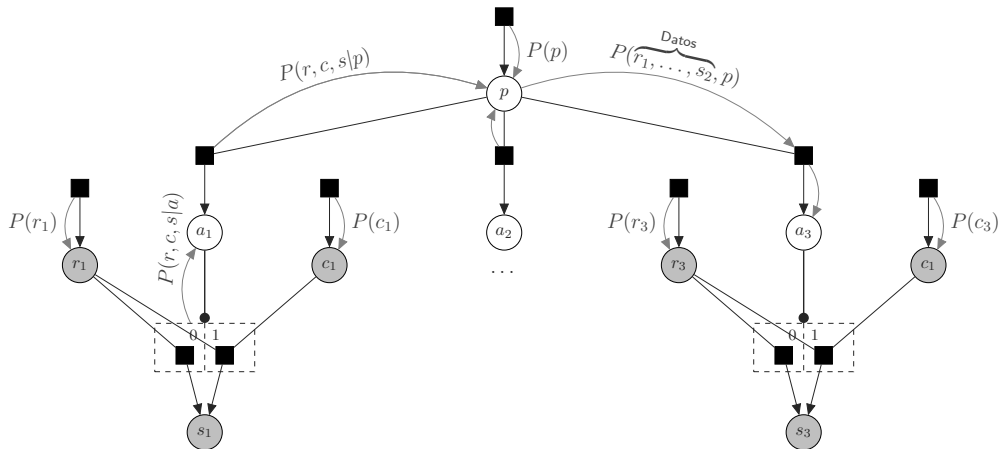
estructuras causales dinámicas



$$m_{p \rightarrow f_{a_{T+1}}}(p) = P(p) \prod_{i=1}^T P(r_i, c_i, s_i|p) = P(r_1, c_1, \dots, c_T, s_T, p)$$

Flujo de inferencia

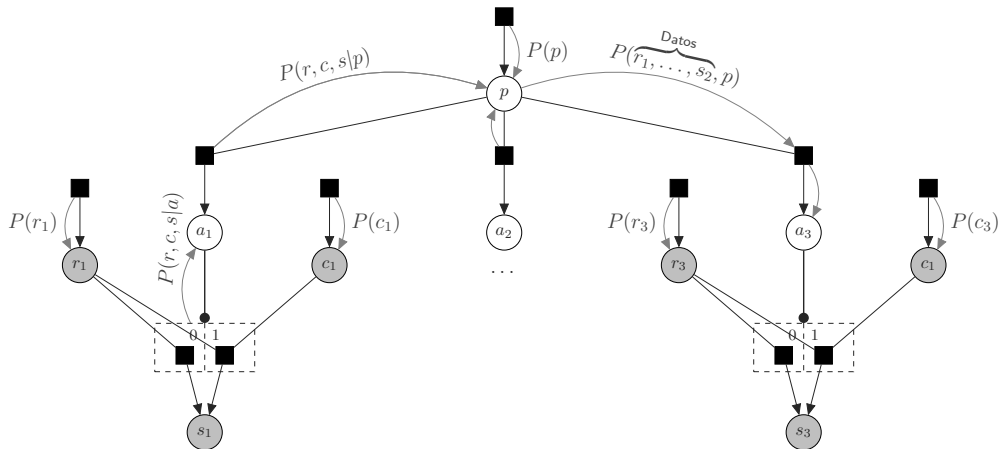
estructuras causales dinámicas



$$m_{f_{a_3 \rightarrow a_3}}(a_3) =$$

Flujo de inferencia

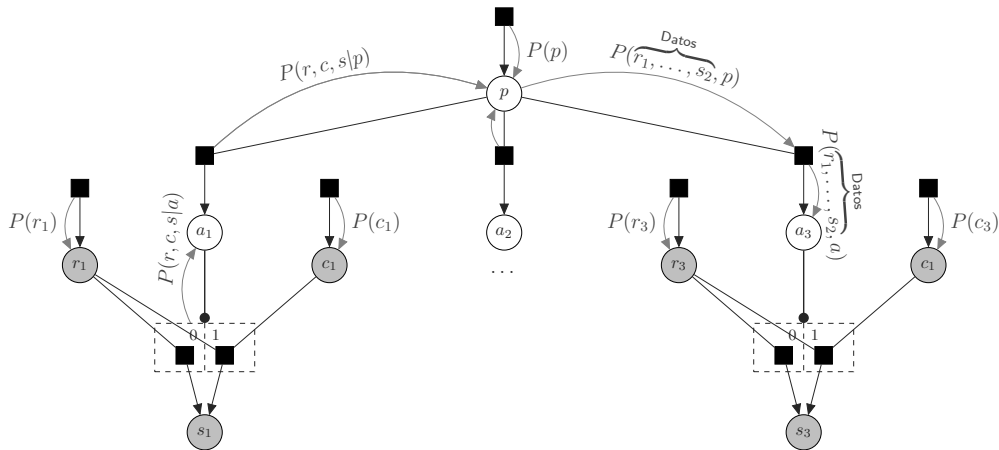
estructuras causales dinámicas



$$m_{f_{a_3 \rightarrow a_3}}(a_3) = \sum_p P(a_3|p)P(r_1, c_1, s_1, r_2, c_2, s_2, p)$$

Flujo de inferencia

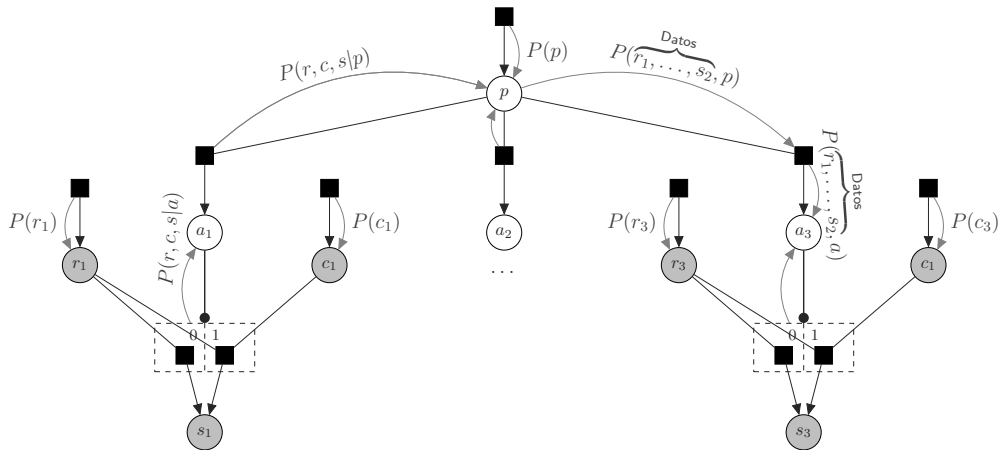
estructuras causales dinámicas



$$m_{f_{a_3 \rightarrow a_3}}(a_3) = \sum_p P(a_3|p)P(r_1, c_1, s_1, r_2, c_2, s_2, p) = P(r_1, \dots, s_2, a)$$

Flujo de inferencia

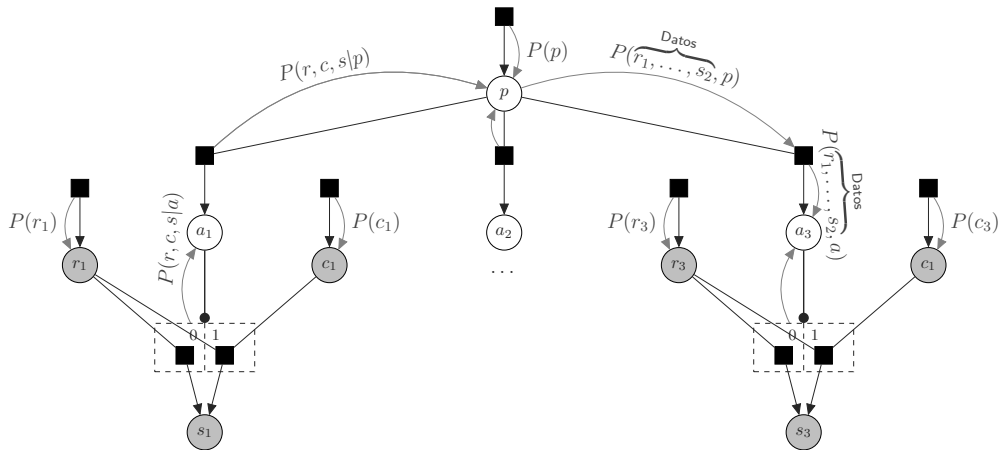
estructuras causales dinámicas



$$P(a_3 | \overbrace{r_1, \dots, s_3}^{\text{Datos}}) \propto$$

Flujo de inferencia

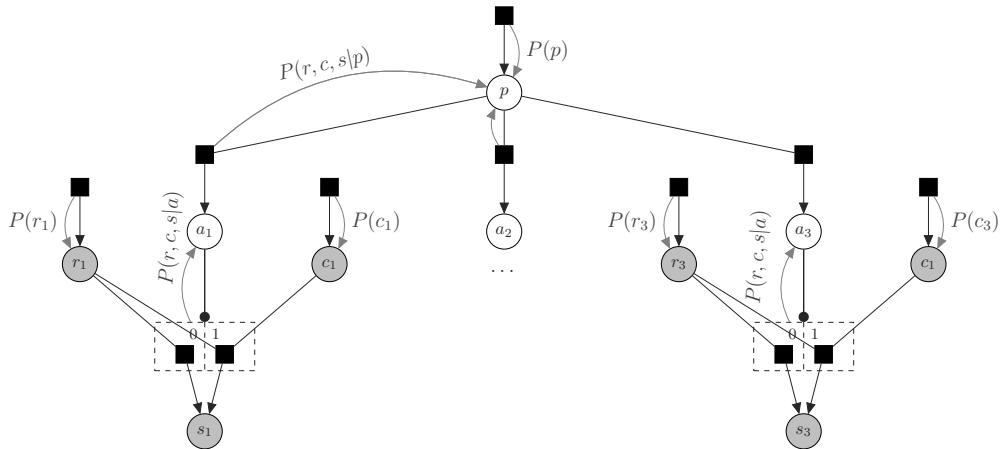
estructuras causales dinámicas



$$P(a_3 | \overbrace{r_1, \dots, s_3}^{\text{Datos}}) \propto P(r_1, c_1, s_1, r_2, c_2, s_2, a_3) P(r_3, c_3, s_3 | a_3)$$

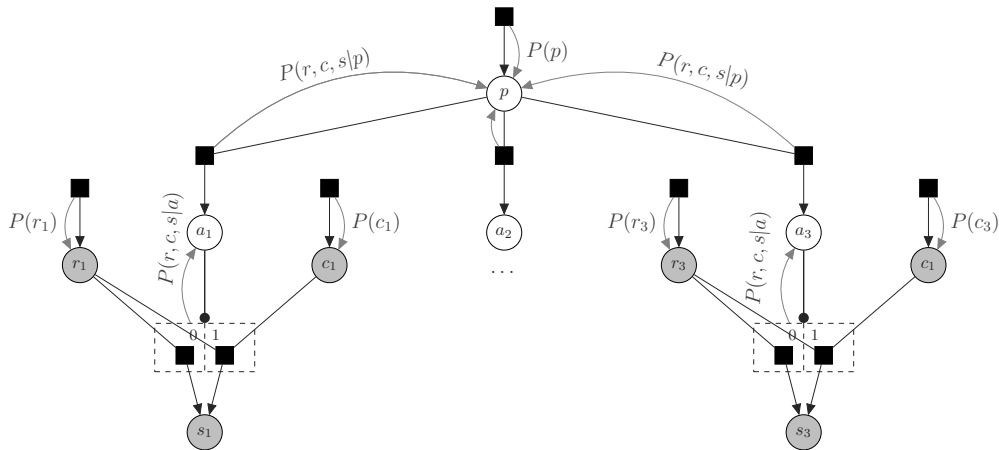
Flujo de inferencia

estructuras causales dinámicas



Flujo de inferencia

estructuras causales dinámicas



$$P(p1, \overbrace{r_1, \dots, s_3}^{\text{Datos}}) = P(p) \prod_{i=1}^T P(r_i, c_i, s_i|p)$$

Analizando el likelihood

$$P(r, c, s|p) = \sum_a P(r)P(c)P(s|r, c, a)P(a|p)$$

Analizando el likelihood

$$P(r, c, s|p) = \sum_a P(r)P(c)P(s|r, c, a)P(a|p) = \underbrace{P(r)P(c)}_{\frac{1}{3} \frac{1}{3}} \left(P(s|r)(1-p) + P(s|r, c)p \right)$$

Analizando el likelihood

$$P(r, c, s|p) = \sum_a P(r)P(c)P(s|r, c, a)P(a|p) = \underbrace{P(r)P(c)}_{\frac{1}{3} \frac{1}{3}} \left(P(s|r)(1-p) + P(s|r, c)p \right)$$

Caso 0: $r = c$. No podemos saber si se olvidó o se acordó.

$$P(s|r) = P(s|r, c) = 1/2 \implies P(r, c, s|p) = 1/18$$

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Caso 1: $r \neq c$ y $c = s$. Seguro se olvidó.

$$P(s|r) = 1/2, P(s|r, c) = 0 \implies P(r, c, s|p) = (1-p)/18$$

Analizando el likelihood

$$P(r, c, s|p) = \sum_a P(r)P(c)P(s|r, c, a)P(a|p) = \underbrace{P(r)P(c)}_{\frac{1}{3} \frac{1}{3}} \left(P(s|r)(1-p) + P(s|r, c)p \right)$$

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$$P(s|r) = 1/2, P(s|r, c) = 0 \implies P(r, c, s|p) = (1-p)/18$$

Caso 2: $r \neq c$ y $c \neq s$. Quizás no se olvidó.

$$P(s|r) = 1/2, P(s|r, c) = 1 \implies P(r, c, s|p) = (1+p)/18$$

Analizando el likelihood

$$P(r, c, s|p) = \sum_a P(r)P(c)P(s|r, c, a)P(a|p) = \underbrace{P(r)P(c)}_{\frac{1}{3} \frac{1}{3}} \left(P(s|r)(1-p) + P(s|r, c)p \right)$$

Caso 0: $r = c$. No podemos saber si se olvidó o se acordó.

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Caso 1: $r \neq c$ y $c = s$. Seguro se olvidó.

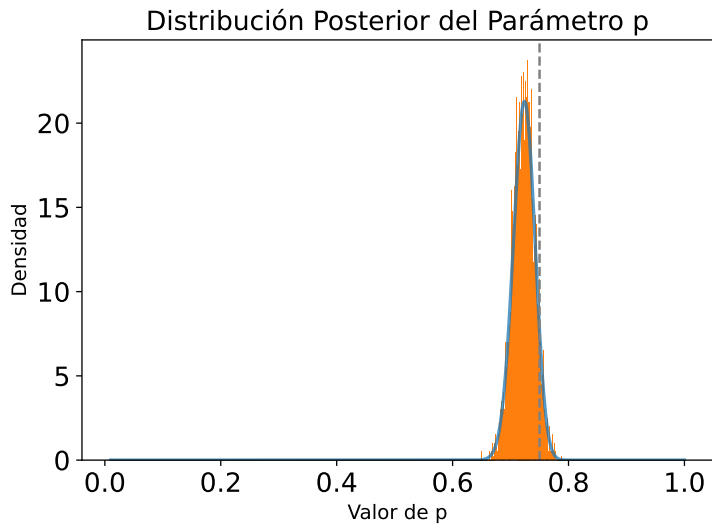
$$P(s|r) = 1/2, P(s|r, c) = 0 \implies P(r, c, s|p) = (1-p)/18$$

Caso 2: $r \neq c$ y $c \neq s$. Quizás no se olvidó.

$$P(s|r) = 1/2, P(s|r, c) = 1 \implies P(r, c, s|p) = (1+p)/18$$

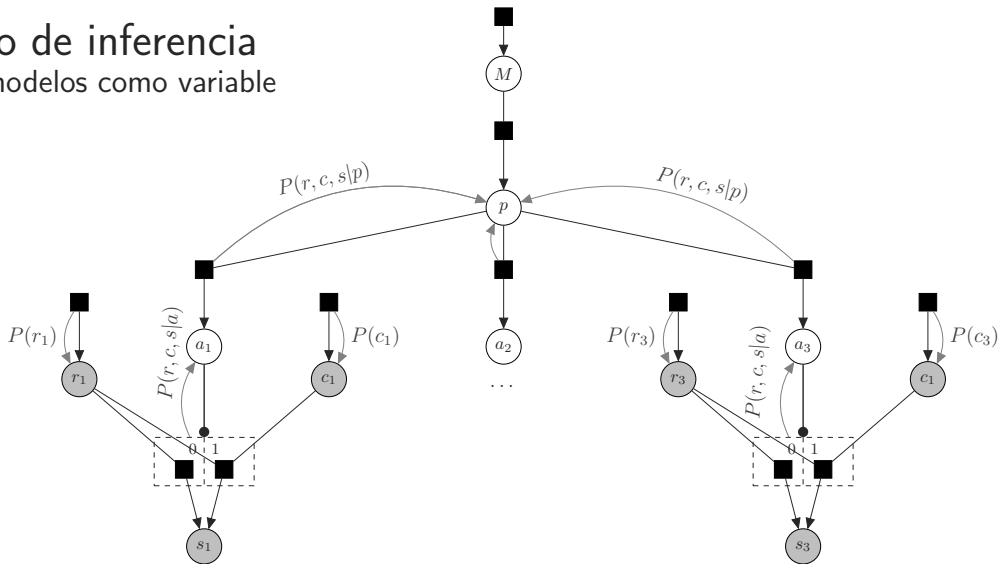
$$\prod_{t=1}^T P(r_t, c_t, s_t|p) = \frac{1}{18} \left(1^{n_0} (1-p)^{n_1} p^{n_2} \right)$$

Analizando el likelihood



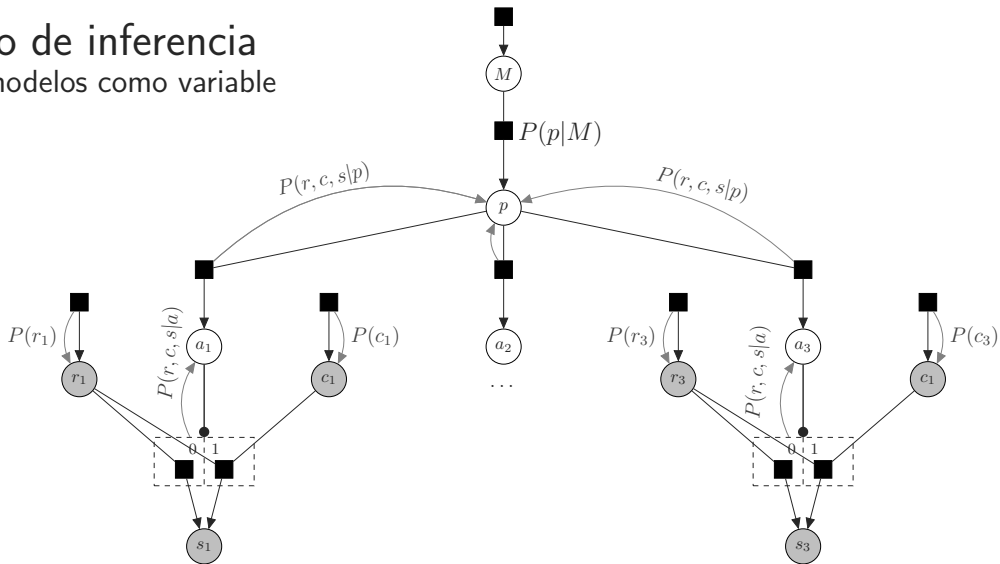
Flujo de inferencia

Los modelos como variable



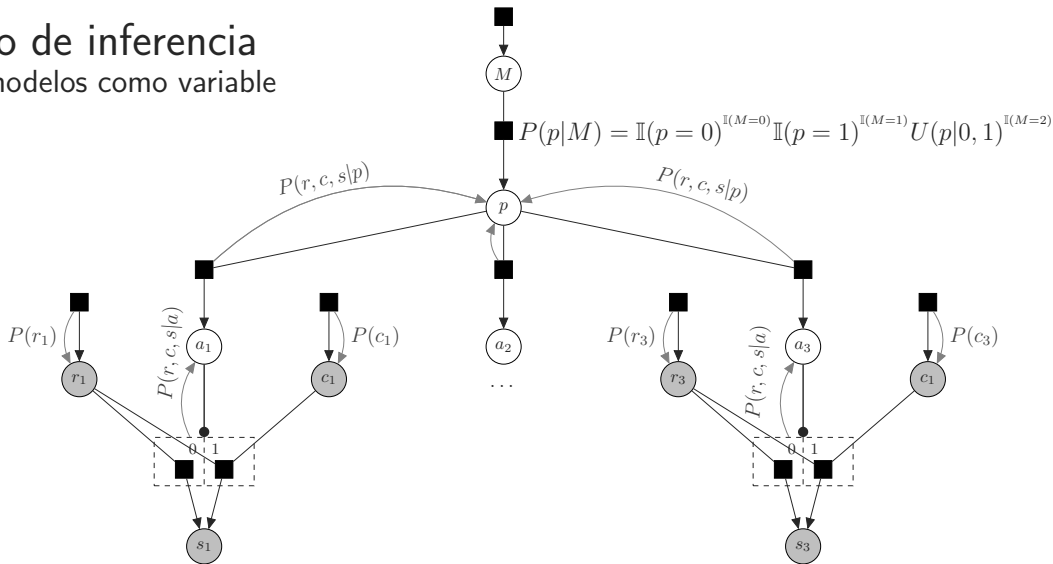
Flujo de inferencia

Los modelos como variable



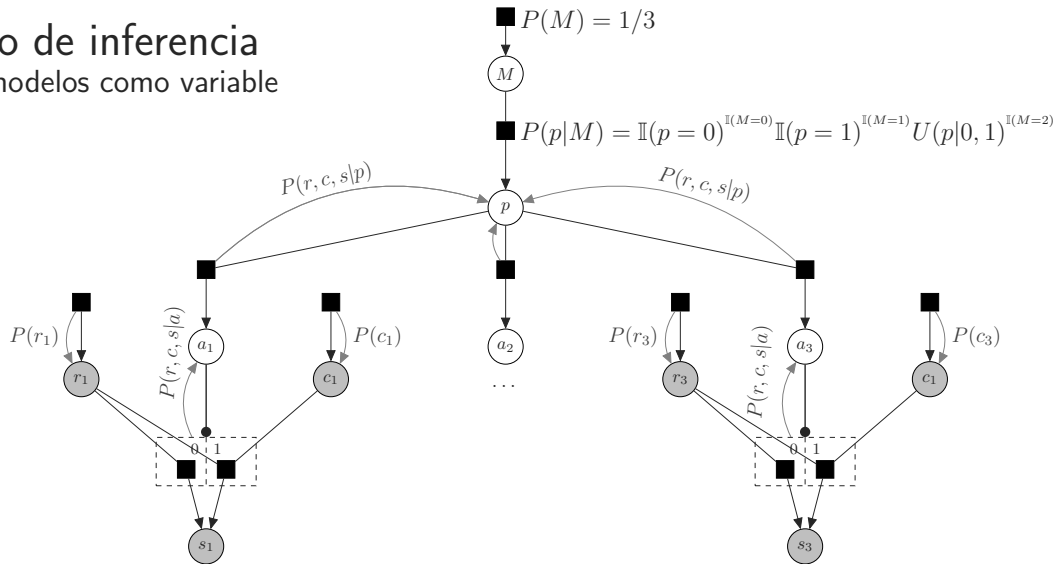
Flujo de inferencia

Los modelos como variable



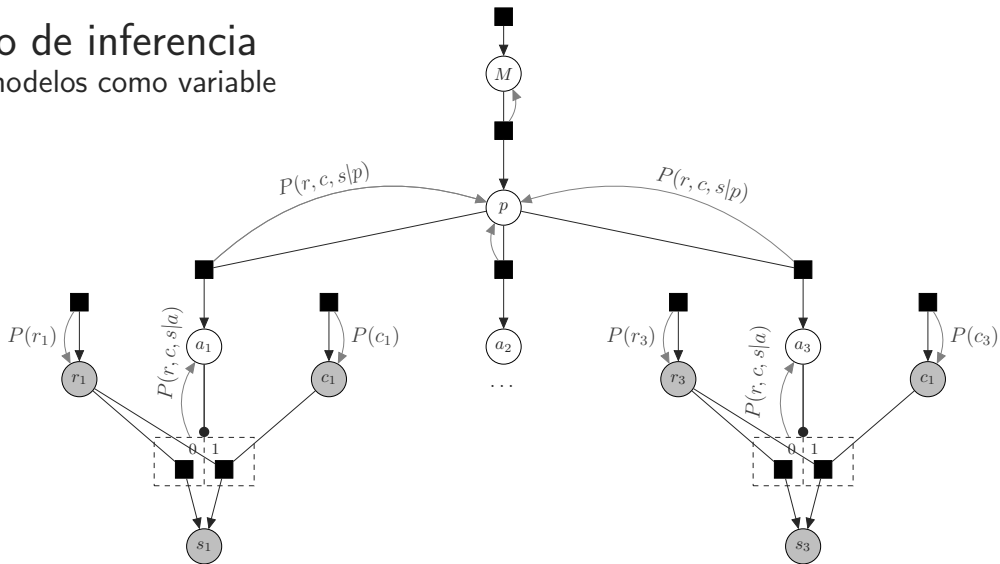
Flujo de inferencia

Los modelos como variable



Flujo de inferencia

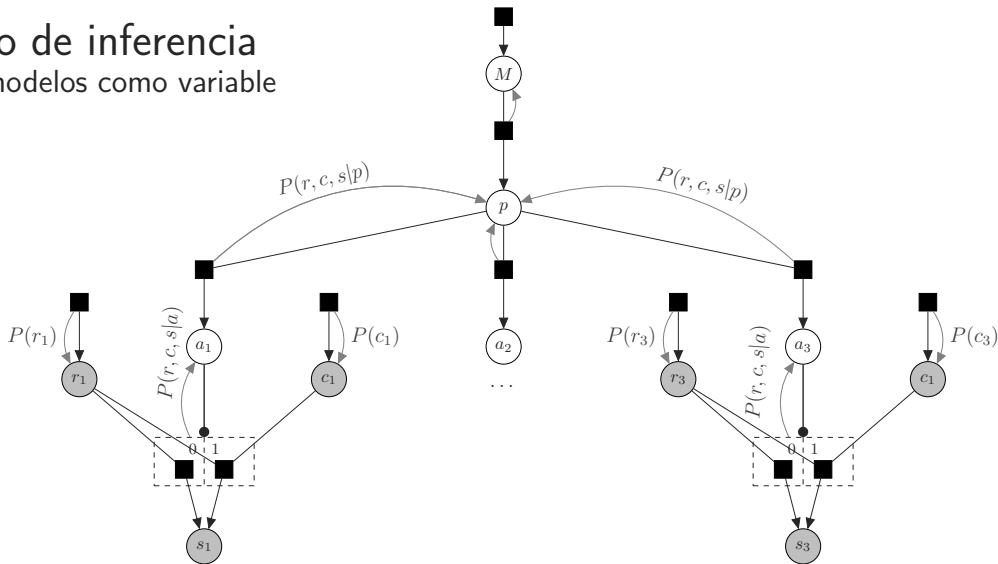
Los modelos como variable



$$m_{f_p \rightarrow M}(M) =$$

Flujo de inferencia

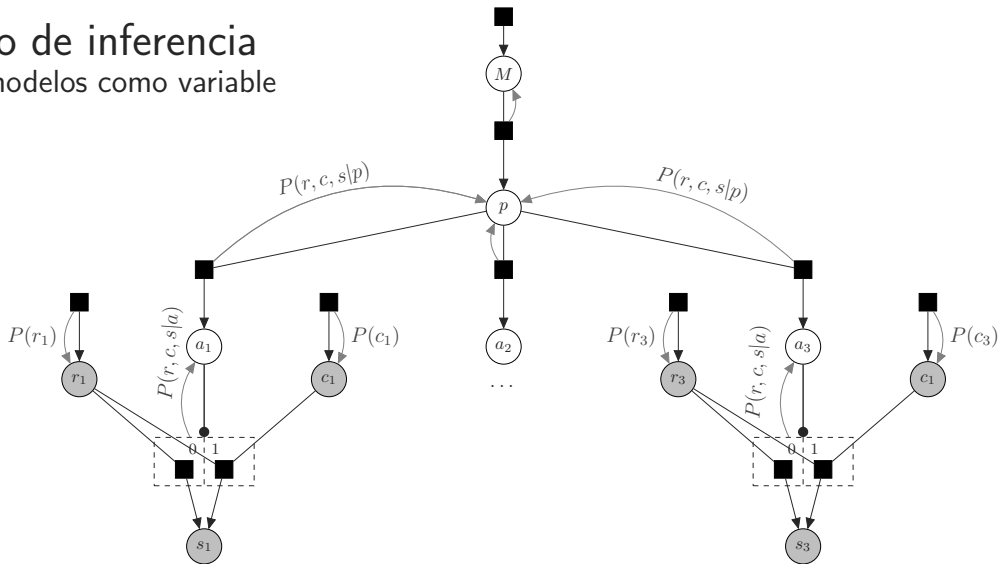
Los modelos como variable



$$m_{f_p \rightarrow M}(M) = \sum_p P(p|M) \prod_{i=1}^T P(r_i, c_i, s_i|p)$$

Flujo de inferencia

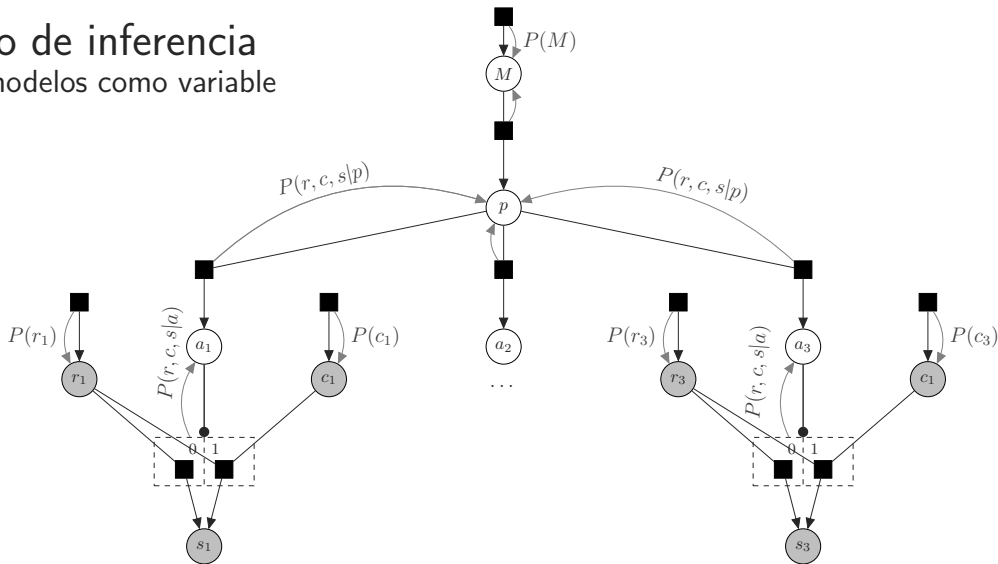
Los modelos como variable



$$m_{f_p \rightarrow M}(M) = \sum_p P(p|M) \prod_{i=1}^T P(r_i, c_i, s_i|p) = P(\text{Datos}|M)$$

Flujo de inferencia

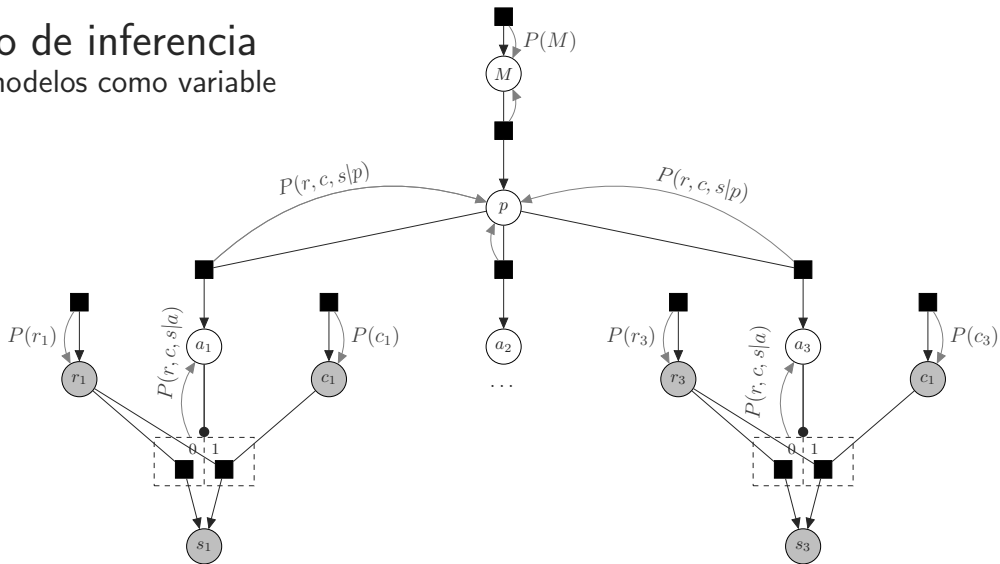
Los modelos como variable



$$m_{f_p \rightarrow M}(M) = \sum_p P(p|M) \prod_{i=1}^T P(r_i, c_i, s_i|p) = P(\text{Datos}|M)$$

Flujo de inferencia

Los modelos como variable



$$P(M, \text{Datos}) = P(\text{Datos}|M)P(M)$$

$p = \mathbf{b}$

Laboratorios de
Métodos Bayesianos

Bibliografía Semana 3

- Winn 2019. *Model-Based Machine Learning*. Cambridge university press; 2003 ([Descargar](#)). **Lectura: cap 3.**

Otras:

- Herbrich, Minka, Graepel. 2006. *TrueSkill: A Bayesian Skill Rating System* ([Descargar](#)). **Lectura: paper.**